NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

School of Electrical & Electronic Engineering

EE6108 Computer Networks

Academic Year 2025-2026

L6108

Enterprise Switched LAN Design and Implementation

Communication Research Laboratory (S2-B4c-17)

Dress Code in the Laboratory

- **Work shirt that covers the upper torso and arms.**
- **Lower body clothing that covers the entire leg.**
- **Closed-toe shoes that cover the top of the foot.**

Please access this link:
http://tiny.cc/f6f5001 **or scan the QR code below to access EEE CA Guidelines.**
Request for MAKE-UP LAB SESSION must be submitted only after the online leave application form has been submitted.

Laboratory Manual

NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING

ACADEMIC YEAR 2025/26

L6108 LABORATORY MANUAL

Enterprise Switched LAN Design and Implementation

Communication Lab
(S2 – B4c – 17)

Students are expected to read through all the theoretical parts of this manual before attending this laboratory session

NAME : ____

GROUP : ____

DATE : ____

INSTRUCTION

1. This 1.5 tadays laboratory session consists of three main parts:

   Part I: IP Networking Fundamentals including Subnetting, Routing and Switching
   Part II: Spanning Tree Protocol (STP) and Layer 2 Loop Prevention
   Part III: VLAN Configuration, EtherChannel, and Inter-VLAN Routing
   Part IV: Network Address Translation(NAT)and External Communication

2. Enter your experimental results only in the space provided in this manual
3. Students may ask the lecturer-in-charge questions concerning the theory, laboratory equipment and experiment procedures but not questions set out in this manual.
4. Keep your answers short so that they can be written within the space provided in this manual.
5. Make sure you understand what you are doing and know how to operate the instruments. If you don't, please ask.

Part 1. IP Networking Fundamentals

1. Objectives

- To familiarize with Cisco Packet Tracer software interface and build a physical network topology independently.
- To perform IP Subnetting using standard formulas to divide a Class C network into multiple functional subnets.
- To study and understand the principles of Layer 2 Switching and Multi-link Physical Routing to enable cross-subnet communication.
- To configure DHCP (Dynamic Host Configuration Protocol) and utilize network troubleshooting skills to resolve broadcast domain conflicts.

2. Preparation

- Cisco Packet Tracer (Latest Version)
- 1 switch (Like 2960).
- 3 PC.
- 1 router (Like 2911).

3. Experiment 1 -- Topology Setup

a. Build the network topology in Packet Tracer

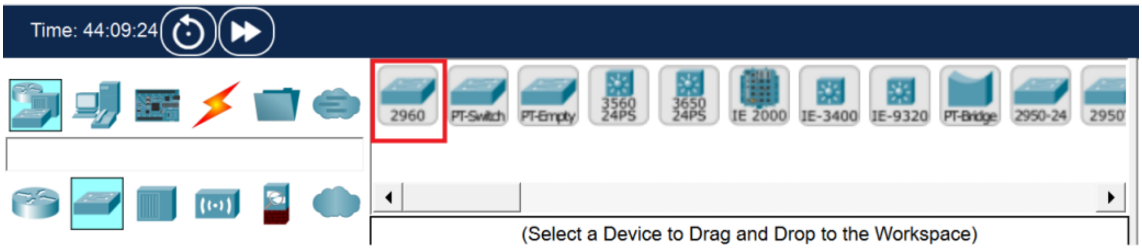
Please complete the physical topology connection according to the instructions below:

i. Open Cisco Packet Tracer.

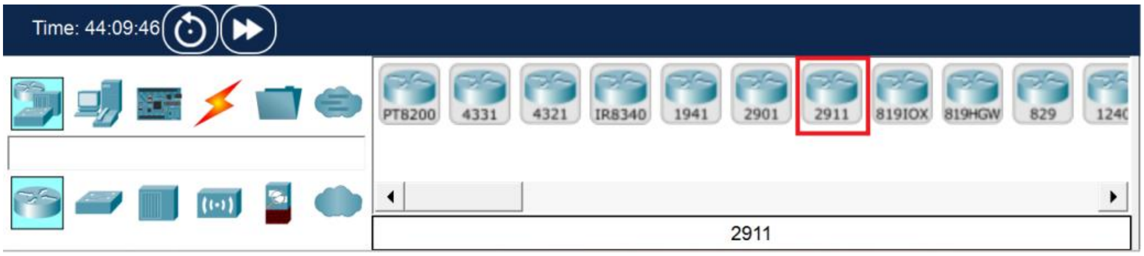
ii. In the lower-left corner of the Packet Tracer window, locate the Device-Type Selection Box.

iii. Click End Devices, drag PC into the workspace, and place 3 PCs. Rename them as: PC0, PC1, and PC2.

iv. Click Network Devices, then select Switches. Drag one 2960 Switch into the workspace.

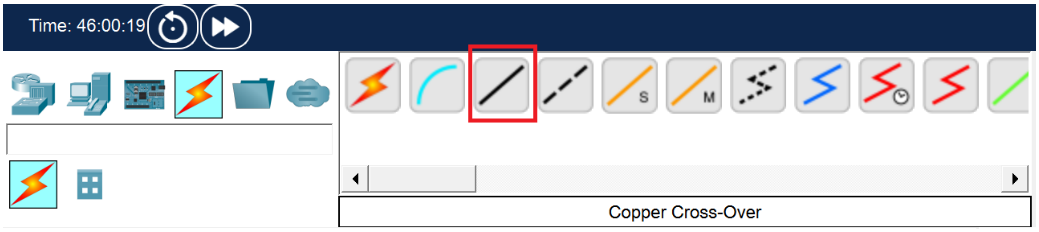


v. Click Network Devices, then select Routers. Drag one 2911 Router (which provides 3 GigabitEthernet interfaces) and place it above the switch.



b. Cabling the Network

i. Click Connections (the lightning icon) and select Copper Straight-Through cable.



ii. Connect the devices according to the following tables to establish the 3 independent physical link:

Device	**Interface**	**Connected Device**	**Interface**
PC0	FastEthernet0	Switch0	FastEthernet0/1
PC1			FastEthernet0/2
PC2			FastEthernet0/3

Switch0	FastEthernet0/4	Router0	GigabitEthernet0/0
	FastEthernet0/5		GigabitEthernet0/1
	FastEthernet0/6		GigabitEthernet0/2

Note: The link lights on the router interfaces will initially appear as red triangles because the ports are administratively down by default. They will be activated in Experiment 3.

4. Experiment 2 -- IP Subnetting

In this experiment, the base network address is given as 192.168.10.0/24. You are required to design a network for 3 colleges of NTU (EEE, NIE, CCDS) , each requiring approximately 20 hosts.

Calculate the required host bits(n) using the formula: $2^n - 2 \geq required\ hosts$.

a. Calculation & Thinking

i. In the host calculation formula $2^n - 2 \geq required\ hosts$, why must we subtract 2 from the total permutations?

Because in any IPv4 subnet, two specific IP addresses are strictly reserved and cannot be assigned to any host computer or router interface:

- Network Address: Where all host bits are 0, used to identify the subnet itself.
- Broadcast Address: Where all host bits are 1, used to send broadcast messages simultaneously to all devices within that specific subnet.

ii. Based on the requirement of 20 hosts per college, calculate the required host bits (n) and derive the new subnet mask (please provide both CIDR notation and decimal format).

- Calculate host bits: Using the formula, we get n=5.
- Derive the mask: The total length of an IPv4 address is 32 bits. 32 - 5=27 network bits.
- Final Result: The CIDR notation is /27, and the decimal format is 255.255.255.224.

iii. Taking NIE as an example, if its starting network address is 192.168.10.32, what is its "Broadcast Address" based on the subnet size calculated in Q2? Can PC1 in NIE configure this broadcast address on its own network interface?

The broadcast address for NIE is 192.168.10.63.

Explanation: The subnet block starts from .32 and contains exactly 32 IPs. The last IP in this block is .63. PC1 cannot use this address. If configured, the operating system will reject it because it is strictly reserved for network broadcasting.

b. Complete the IP Allocation Table:

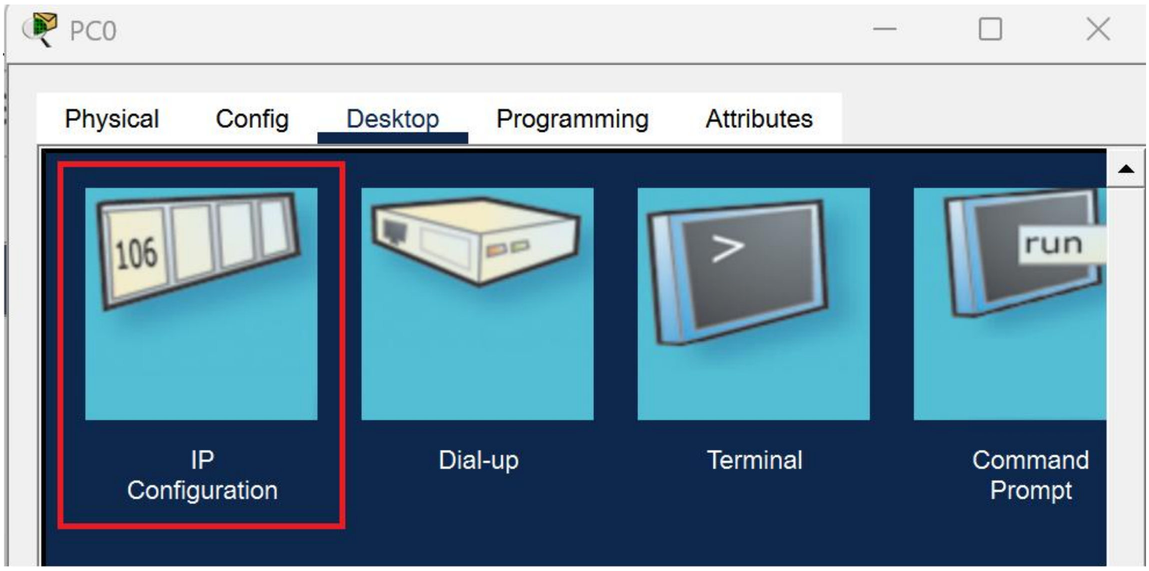
College	Network Address	Usable IP Range	Subnet Mask	Default Gateway
EEE (PC0)	192.168.10.0	192.168.10.1-.30	255.255.255.244	192.168.10.1
NIE (PC1)	192.168.10.32	192.178.10.33-.62	255.255.255.244	192.168.10.33
CCDS (PC2)	192.168.10.64	192.168.10.65-.94	255.255.255.244	192.168.10.65

5. Experiment 3 -- Switching & Routing Implementation

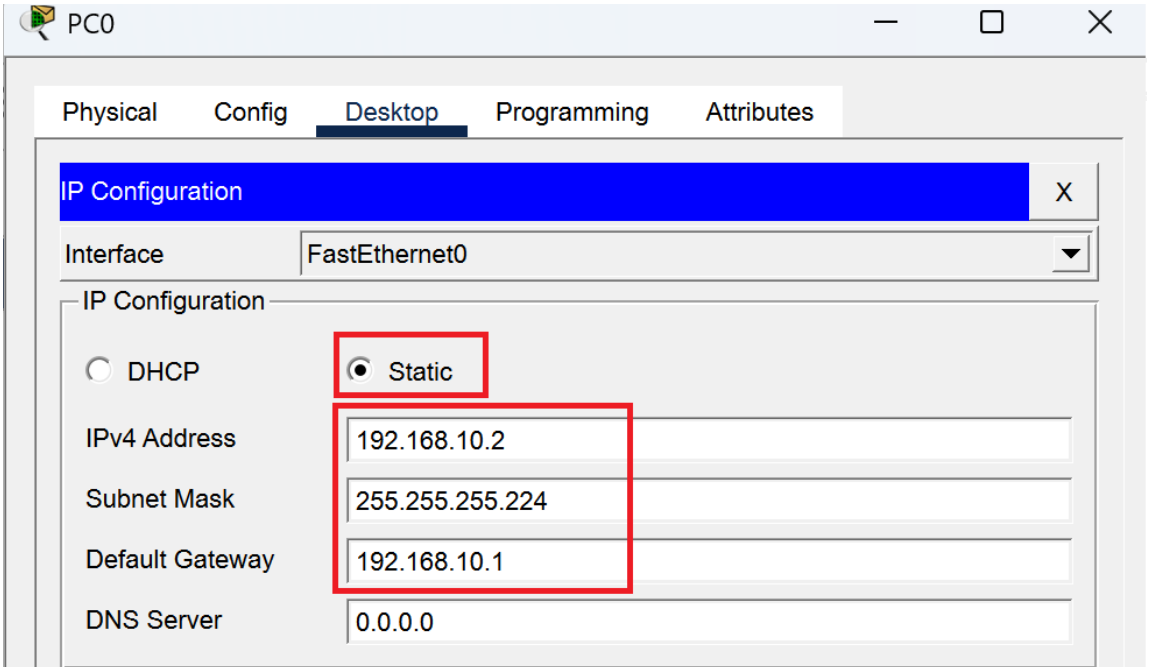
In this experiment, we will configure the end devices and the router to enable inter-subnet communication.

a. Configure PC IP address

i. Click on PC0, navigate to the Desktop tab, and then click on IP Configuration.



ii. Select Static. Set the IPv4 Address (e.g., 192.168.10.2), Subnet Mask (255.255.255.224), and Default Gateway (192.168.10.1) based on your calculation.



Note: When you click on this box, the software will automatically fill in 255.255.255.0. Please make sure to manually delete the last 0 and change it to 224, making it 255.255.255.224. (This is the result of your Experiment 2 !)

iii. Repeat this operation to configure PC1 and PC2 with their respective subnet data.

b. Configure the Layer 2 Switch

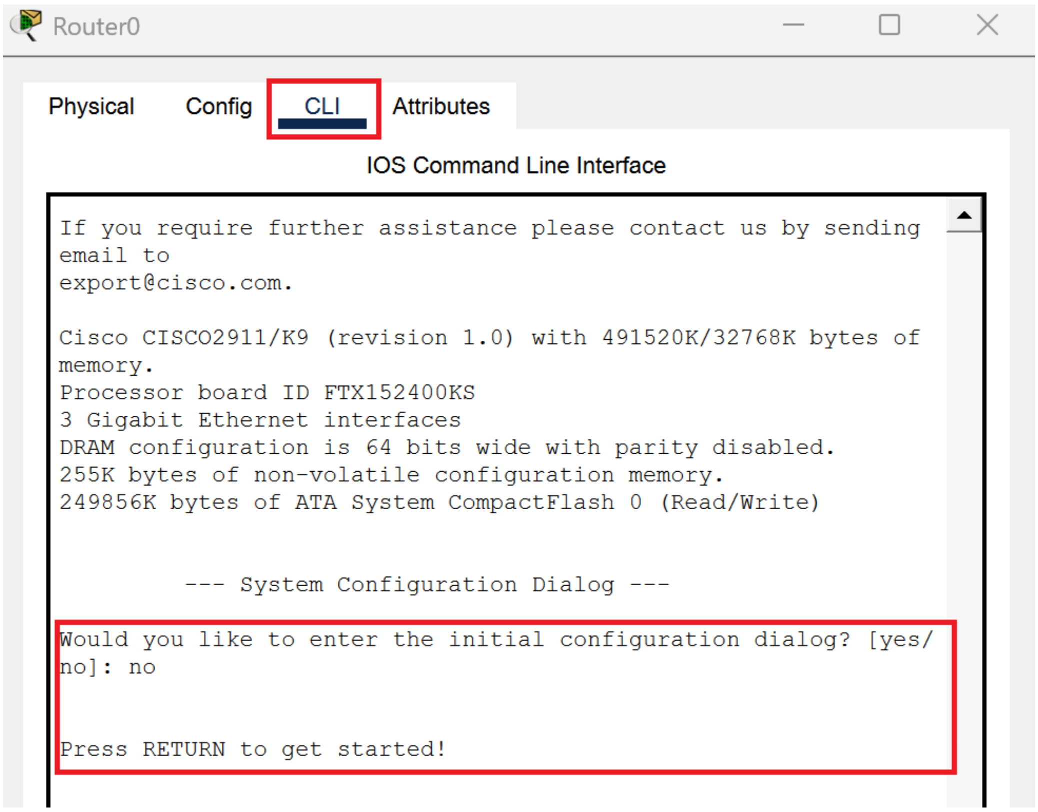
No configuration is required for the 2960 Switch. By default, all 24 interfaces are in the same broadcast domain (VLAN 1). It will automatically build its MAC address table to perform Layer 2 forwarding.

c. Configure the Layer 3 Router

i. Click on PC0, navigate to the Desktop tab, and then click on IP Configuration.

Click on Router0, open the CLI tab.

If you see the prompt ’ Would you like to enter the initial configuration dialog? [yes/no]:’ , type ’ no’ and press Enter.



ii. Press Enter a few times until the Router> prompt appears.

Enter the following commands to activate the interfaces and set the gateways:

Router> **enable**

Router# **configure terminal**

! --- Configure EEE Gateway (G0/0) ---

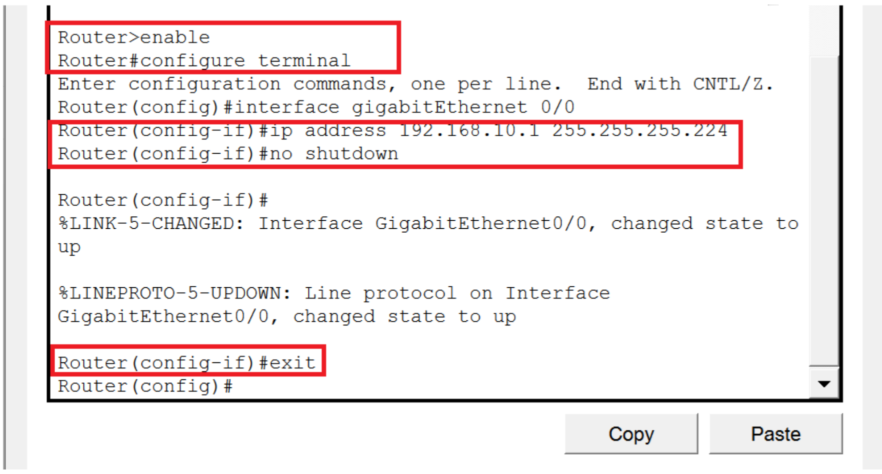
Router(config)# **interface gigabitEthernet 0/0**

Router(config-if)# **ip address 192.168.10.1 255.255.255.224**

Router(config-if)# **no shutdown**

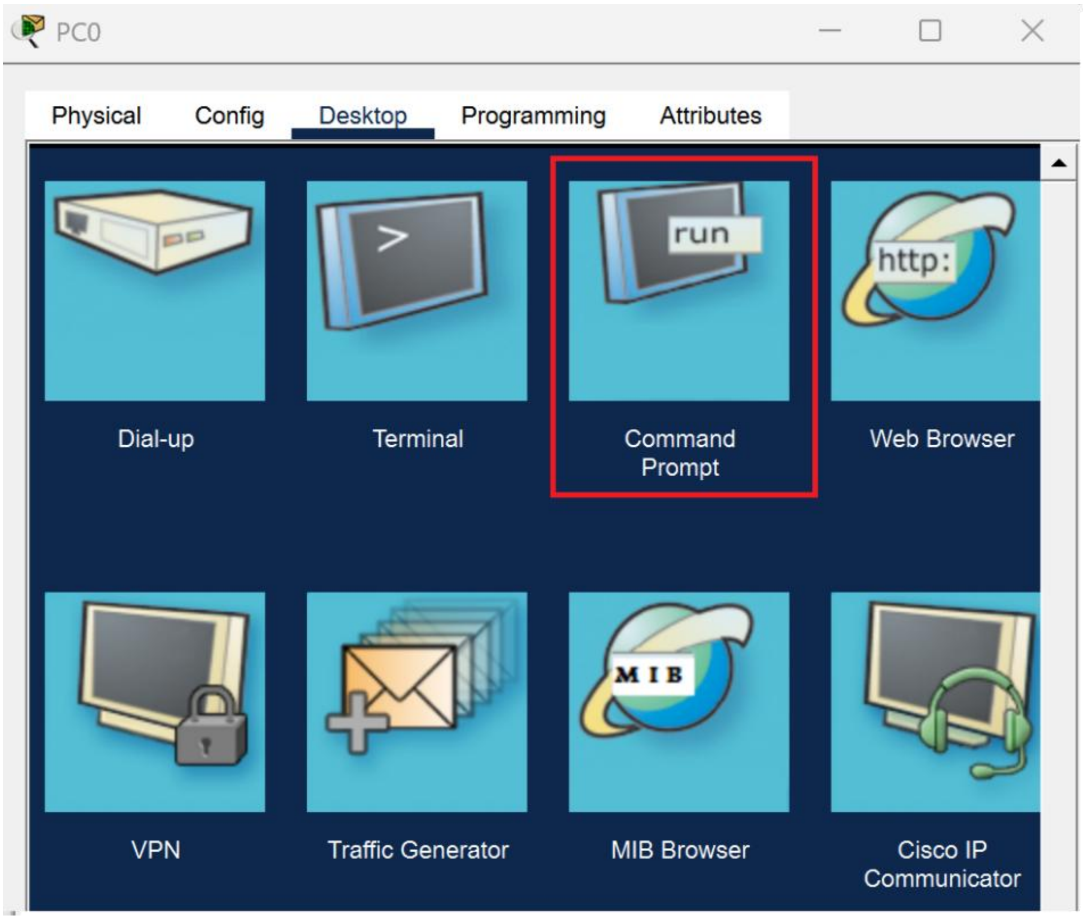
Router(config-if)# **exit**

! (Repeat the same logic to configure G0/1 for NIE and G0/2 for CCDS)

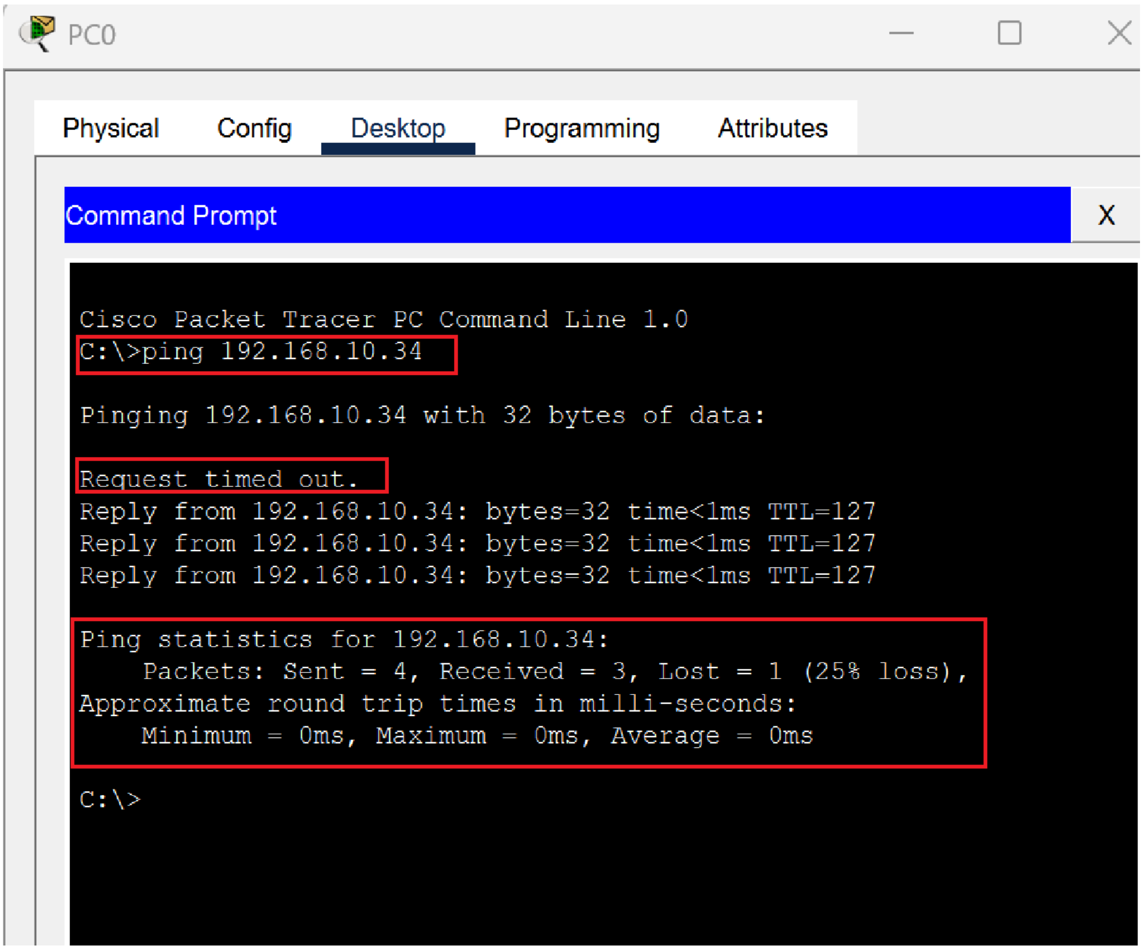


d. Connectivity Verification

i. Open the Command Prompt on PC0.



ii. Enter the command’ ping 192.168.10.34’ to test reachability to PC1.



Note: The first one or two requests may display "Request timed out" due to the ARP resolution process. This is a normal phenomenon. A subsequent "Reply from..." indicates successful routing.

6. Experiment 4 — DHCP & Broadcast Chaos

To improve efficiency, we will configure the router to act as a DHCP server. However, implementing this in a pure physical multi-link topology will trigger a network conflict.

a. Configure DHCP Pools

Enter the following commands in the Router CLI:

Router# **configure terminal**

! Exclude gateway IP addresses

Router(config)# **ip dhcp excluded-address 192.168.10.1**

Router(config)# **ip dhcp excluded-address 192.168.10.33**

Router(config)# **ip dhcp excluded-address 192.168.10.65**

! Create EEE DHCP Pool

Router(config)# **ip dhcp pool Pool_EEE**

Router(dhcp-config)# **network 192.168.10.0 255.255.255.224**

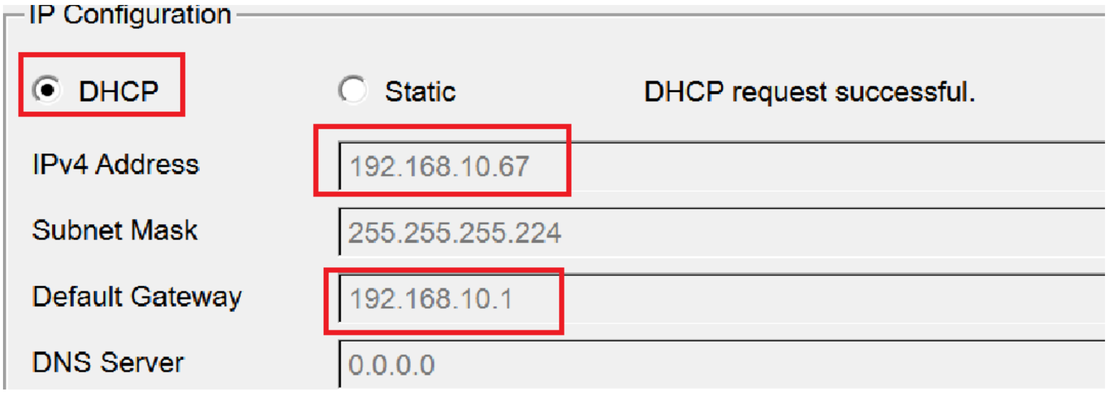
Router(dhcp-config)# **default-router 192.168.10.1**

Router(dhcp-config)# **exit**

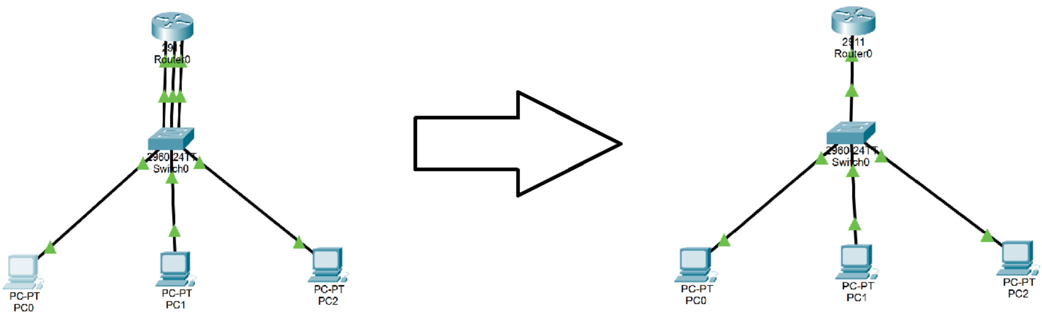
! (Proceed to create Pool_NIE and Pool_CCDS...)

b. Observe the Bug & Physical Verification

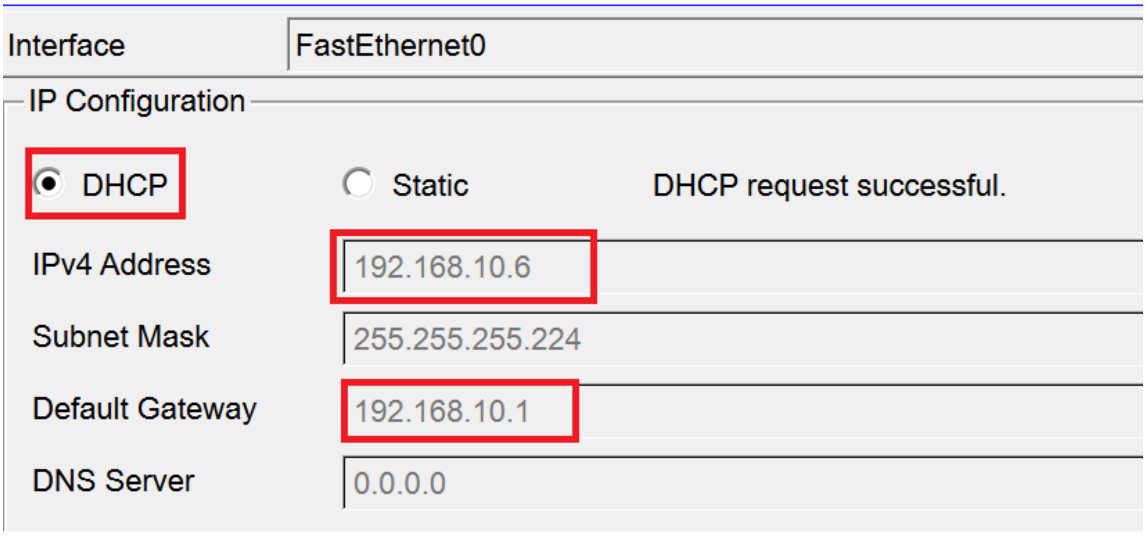
i. On PC0, switch the IP Configuration mode from Static to DHCP. Observe the obtained IP address and Gateway. (You will notice PC0 receives an incorrect IP and Gateway pairing).



ii. Select the Delete tool in Packet Tracer. Remove two of the physical links between the switch and the router, keeping only the G0/0 link.



iii. Switch PC0 back to Static and then to DHCP again. Observe the successful allocation.



c. Practice & Thinking Question

Based on the concept of "Broadcast Domain", explain why the DHCP allocation fails when 3 links are connected, but succeeds when 2 links are removed?

3 Links Connected: By default, the entire 2960 switch operates in a single Broadcast Domain. A DHCP Discover message from a PC is a broadcast. It floods out of all switch ports, hitting all 3 router interfaces simultaneously. All 3 router interfaces respond with IPs from different pools, causing a "Race Condition" and IP mismatch.

2 Links Removed: Removing the cables physically isolates the broadcast. Only one specific router interface (the intended gateway) receives the broadcast and replies correctly.

Note: After completing this experiment, reconnect the 2 deleted links using Copper Straight-Through cables and set all PCs back to Static IP mode for the next phase.

7. Transition to Next Parts

By removing the physical cables in Experiment 4, we temporarily solved the DHCP broadcast conflict, but introduced critical engineering challenges:

a. Reliability Crisis:

Relying on a single switch creates a single point of failure. Introducing redundant switches will cause Layer 2 loops. (To be resolved in Part 2. Spanning Tree Protocol -

STP).

b. Scalability Crisis:

Unplugging cables disables other colleges. Managing a network without logical boundaries is unsustainable. (To be resolved in Part 3. Virtual LAN - VLAN).

Part 2. Spanning Tree Protocol (STP) and Layer 2 Loop Prevention

1. Objectives

- To study and understand the basic principles of Spanning Tree Protocol (STP).
- To build a switched network topology and configure IP connectivity between hosts.
- To introduce redundant links between switches and observe how STP prevents Layer 2 loops.
- To analyze STP information and port roles using CLI commands.
- To understand the Root Bridge election process and influence it by modifying bridge priority.
- To observe how STP recalculates the topology and activates a backup path when a link failure occurs.

2. Preparation

- Cisco Packet Tracer (Lastest Version)
- Lab Manual with Student instruction and Packet tracer file used in the workshop

3. Experiment 1 Topology setup

Please complete the terminal device configuration according to the instructions below：

a. Build the network topology in Packet Tracer

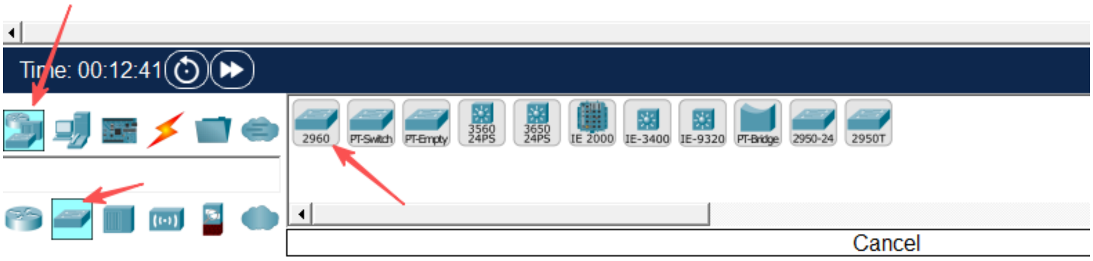
i. Open Cisco Packet Tracer.

ii. In the lower-left corner of the Packet Tracer window, locate the Device-Type Selection Box.

iii. Click Network Devices, then select Switches.

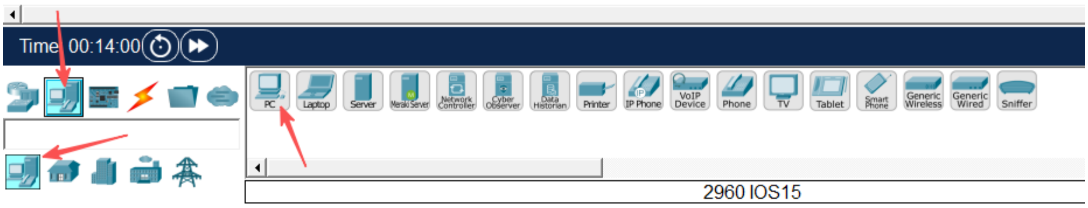
iv. From the device list, drag 2960 Switch into the workspace.

Place two switches and rename them as: Switch0, Switch1



v. In the Device-Type Selection Box, click End Devices.

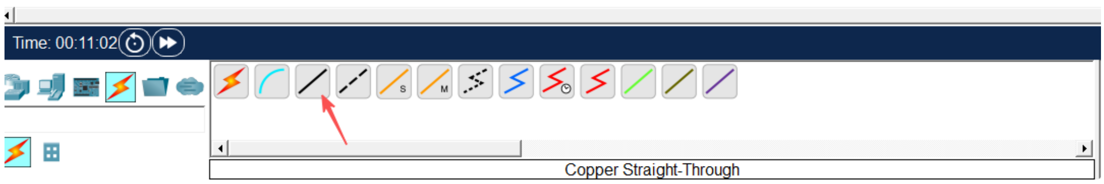
vi. Drag PC into the workspace and place six PCs. Rename them from PC0 to PC5.



vii. Click Connections (the lightning icon).

viii. Select Copper Straight-Through cable and connect the PCs to the switches according to the following table:

Device	Interface	Connected Device	Interface
PC0	FastEthernet0	Switch0	FastEthernet0/1
PC1			FastEthernet0/2
PC2			FastEthernet0/3
PC3		Switch1	FastEthernet0/1
PC4			FastEthernet0/2
PC5			FastEthernet0/3

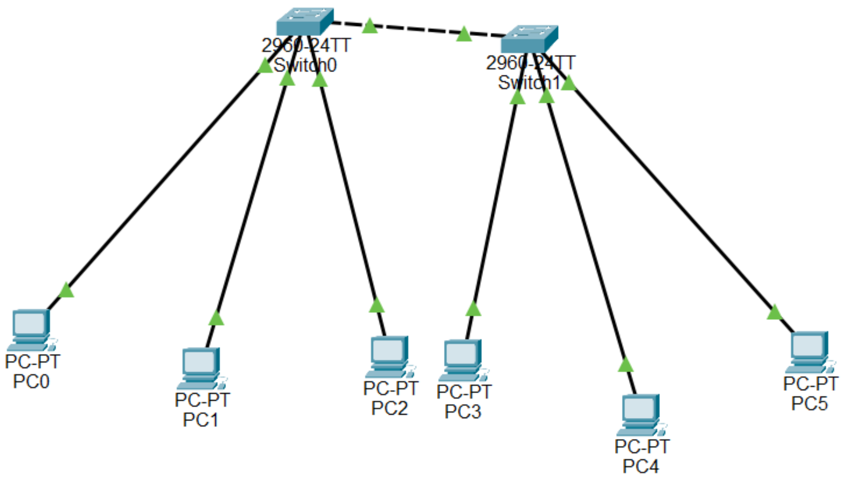


ix. Use Copper cross-over cable to connect the two switches:

Device	Interface	Connected Device	Interface

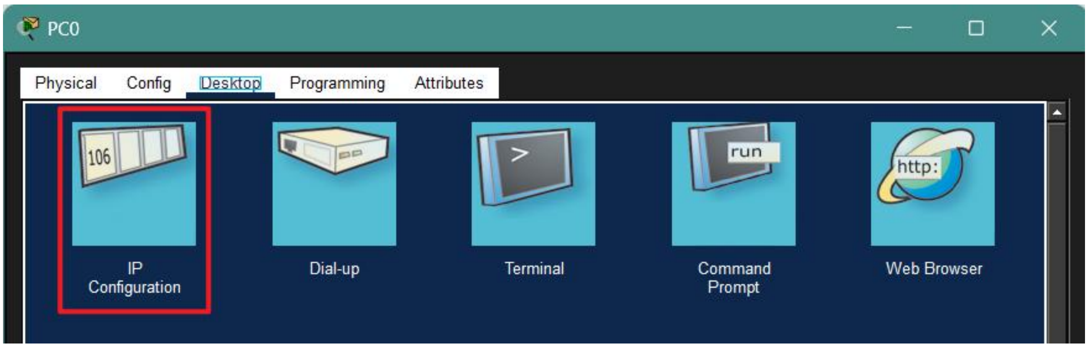
Swtich0	FastEthernet0/4	Swtich1	FastEthernet0/4

After completing the connections, ensure that all link lights turn green, indicating that the connections are active.



b. Configure PC IP address

i. Click on PC0, navigate to the Desktop tab, and then click on IP Configuration.

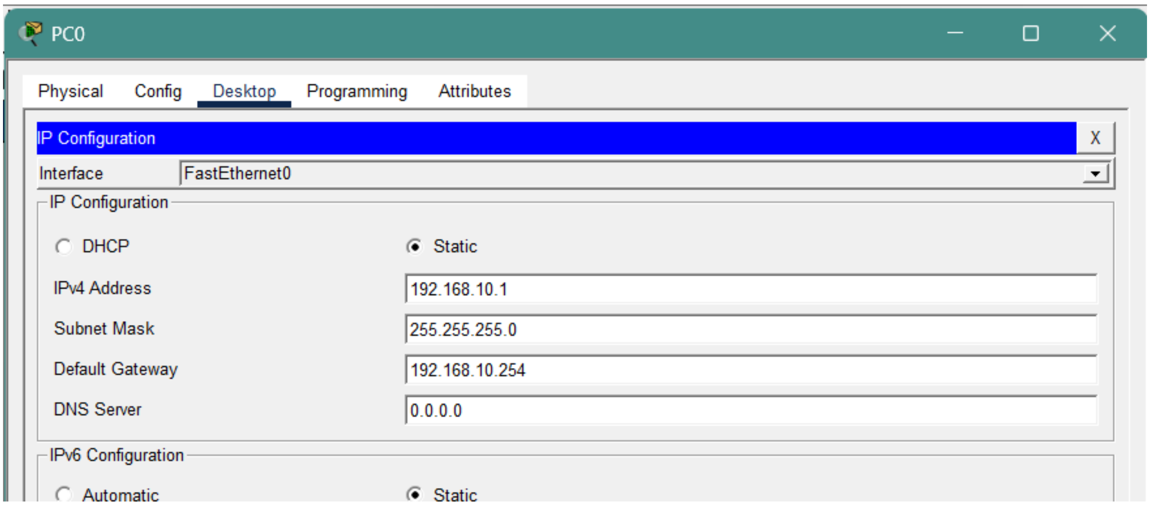


ii. Set the following parameters:

IPv4 Address: 192.168.10.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.254



iii. Repeat this operation to configure PC1-PC5 with IPv4 addresses ranging from 192.168.10.2 to 192.168.10.6.

PC Configuration Reference:

<table>
<tr><th>PC No.</th><th>IPv4 Address</th><th>Subnet Mask</th><th>Default Gateway</th></tr>
<tr><td>PC0</td><td>192.168.10.1</td><td rowspan="4">255.255.255.0</td><td rowspan="4">192.168.10.254</td></tr>
<tr><td>PC1</td><td>192.168.10.2</td></tr>
<tr><td>……</td><td>……</td></tr>
<tr><td>PC5</td><td>192.168.10.6</td></tr>
</table>

Note: You can choose not to follow the table setup. But do configure all six PCs on the same network segment.

4. Experiment 2 Introduce Loop and Observe STP Blocking

In this experiment, an additional physical link will be introduced between the two switches to create a Layer 2 loop.

a. Add an additional link between the switches

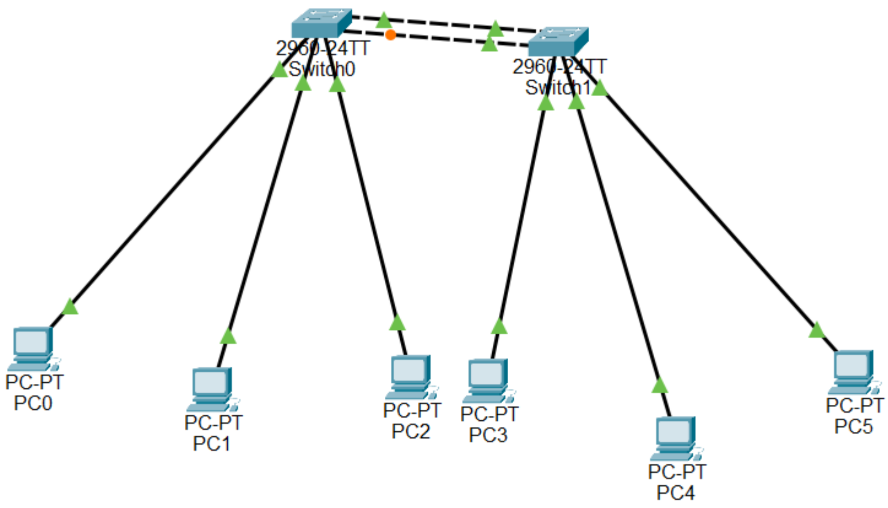
i. In Packet Tracer, click Connections (the lightning icon).

ii. Select Copper cross-over cable.

iii. Connect an additional link between the two switches using the following interfaces:

Device	Interface	Connected Device	Interface

Swtich0	FastEthernet0/5	Swtich1	FastEthernet0/5



After completing the connection, there will now be two physical links between the two switches. Since there are now two parallel links between the switches, a Layer 2 loop is created in the network.

b. Observe the switch port status

When a loop appears in a Layer 2 network, the switch automatically uses the Spanning Tree Protocol (STP) to prevent network loops.

i. Observe the port status indicators on the switch links in Packet Tracer.

ii. Among the two links connecting the switches, one of the ports will change to <u>Orange/Green/Red</u>

iii. This indicates that STP has automatically <u>Blocked/Opened</u> one of the redundant links to prevent a network loop.

c. Verify STP information using CLI

i. Click Switch1 and open the CLI tab.

ii. Enter the following commands:

```
Switch> enable
```

Switch# **show spanning-tree**

iii. From the output, observe the following information:

```
Switch>Enable
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     000C.CFBE.13C5
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000C.CFBE.13C5
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/4            Desg FWD 19        128.4    P2p
Fa0/5            Desg FWD 19        128.5    P2p
```

These values show how STP determines the forwarding path and which port should be blocked.

d. Practice & Thinking Questions

i. Why does one of the switch ports enter the Blocking state after adding a second link?

After a second link is added between the two switches, a redundant Layer 2 path is created. This forms a potential network loop. To prevent frames from circulating endlessly in the loop, Spanning Tree Protocol (STP) automatically places one of the redundant ports into the Blocking state. In this way, only one active forwarding path remains between the switches.

ii. What problems might occur if the network operates without STP?(at least two points)

- Broadcast storm: broadcast frames may circulate endlessly in the network.
- MAC address table instability: switches may keep learning the same MAC address from different ports.
- Duplicate frames: devices may receive multiple copies of the same frame.
- Network congestion or failure: the network may become extremely slow or completely unavailable.

iii. What is the main purpose of Spanning Tree Protocol in a Layer 2 network?

The main purpose of Spanning Tree Protocol (STP) is to prevent Layer 2 switching loops while still allowing redundant physical links in the network. STP does this by automatically blocking unnecessary paths and keeping only one active forwarding path. If the active path fails, STP can recalculate the topology and enable a backup path.

iv. Complete the STP Information Table of Switch0:

Item	**Value**
Root Bridge Priority	32769
Root Bridge MAC Address	The Root Bridge MAC address may vary depending on which switch is elected as the root bridge.
Root Path Cost	19
Root Port	FastEthernet0/4
Designated Ports	Fa0/1, Fa0/2, Fa0/3
Blocking Port	Fa0/5

5. Experiment 3 Root Bridge Election

In this experiment, we will investigate how Spanning Tree Protocol (STP) elects the Root Bridge in a network and observe how changing the bridge priority can influence the election result.

a. Observe the current root bridge

i. Click on Switch0 and open the CLI tab.

ii. Enter the following commands:

Switch> **enable**

Switch# **show spanning-tree**

iii. Repeat the same command on Switch1 and find the root bridge.

b. Change & Verify the bridge priority

STP selects the Root Bridge based on the Bridge ID, which is composed of:

Bridge ID = Priority + MAC Address

A switch with the lowest Bridge ID will become the Root Bridge.

i. On Switch0, enter the following commands:

Switch> **enable**

Switch# **configure terminal**

Switch(config)# **spanning-tree vlan 1 priority 4096**

This command lowers the bridge priority of Switch0, increasing its chance of becoming the Root Bridge.

ii. On Switch0, run the following command again:

Switch# **show spanning-tree**

iii. Check whether the output now shows:

This bridge is the root

iv. Run the same command on Switch1 and observe how the Root Bridge information changes.

c. Practice & Thinking Questions

i. What factor determines which switch becomes the Root Bridge?

The Root Bridge is determined by the Bridge ID, which consists of the bridge priority and the MAC address. The switch with the lowest Bridge ID will be elected as the Root Bridge. If multiple switches have the same priority, the switch with the lowest MAC address becomes the Root Bridge.

ii. After changing the Root Bridge, does the Blocking port change? Explain why.

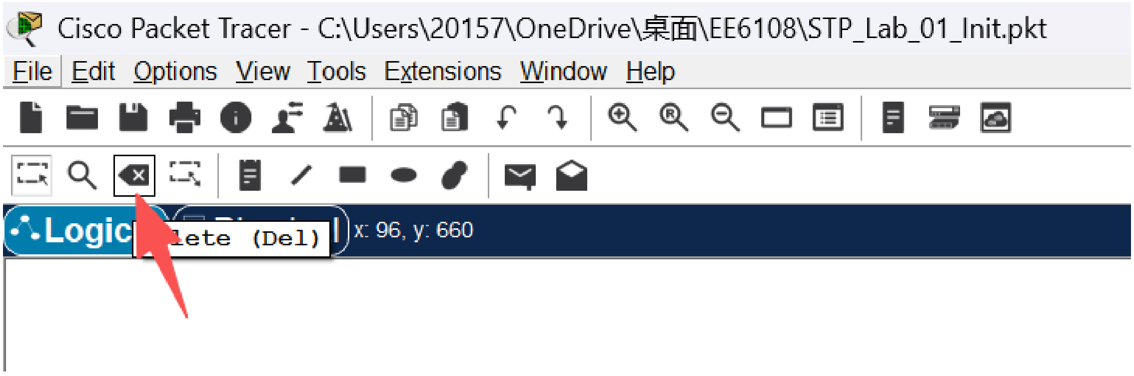
Yes, the Blocking port may change. When the Root Bridge changes, STP recalculates the spanning tree topology. Each switch selects a new Root Port based on the shortest path cost to the new Root Bridge. As a result, the port roles may change and a different port may enter the Blocking state to prevent loops.

6. Experiment 4 Link Failure Recovery

In this experiment, we will observe how Spanning Tree Protocol (STP) responds to a link failure and automatically activates a backup path.

a. Disconnect the active link and observe STP reconvergence

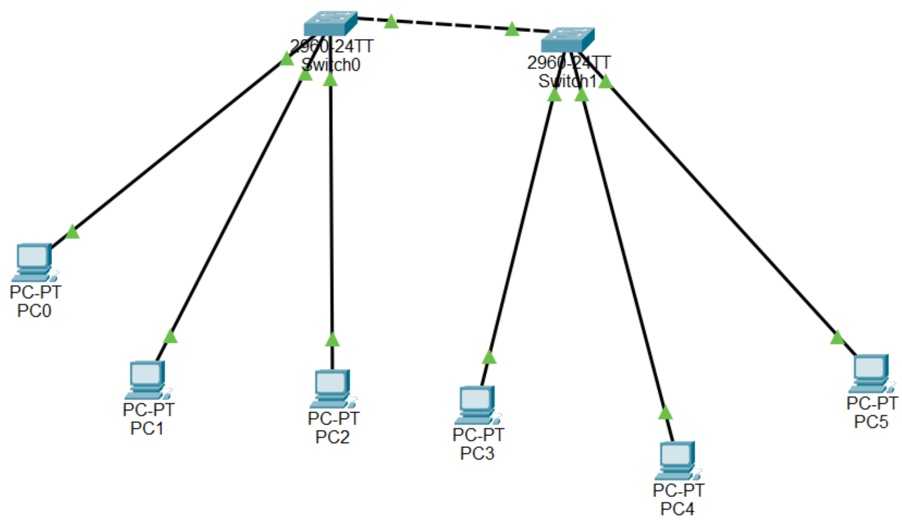
i. In Packet Tracer, select the Delete tool.



ii. Identify the port that is currently in the Blocking state.

iii. Remove the currently active link between the two switches. This link is the one whose ports are in the Forwarding state (Green).

iv. Identify the port that was previously in the Blocking state. After a short period of time, the port state will change from Blocking (Orange) to Forwarding (Green).



v. Click Switch0 and open the CLI tab.

vi. Enter the following command to view the STP information:

Switch> **enable**

Switch# **show spanning-tree**

vii. Observe the changes in the port roles and port states.

b. Practice & Thinking Questions

i. What happens to the previously blocked port after the active link fails?

The previously blocked port transitions from Blocking to Forwarding state. STP recalculates the network topology and activates the backup path to maintain network connectivity.

ii. What is the advantage of having redundant links in a network?

Redundant links provide network reliability and fault tolerance. If one link fails, another link can automatically take over, ensuring continuous communication in the network.

Part 3. Virtual LAN(VLAN)Management, Link Aggregation, and Inter-VLAN Routing

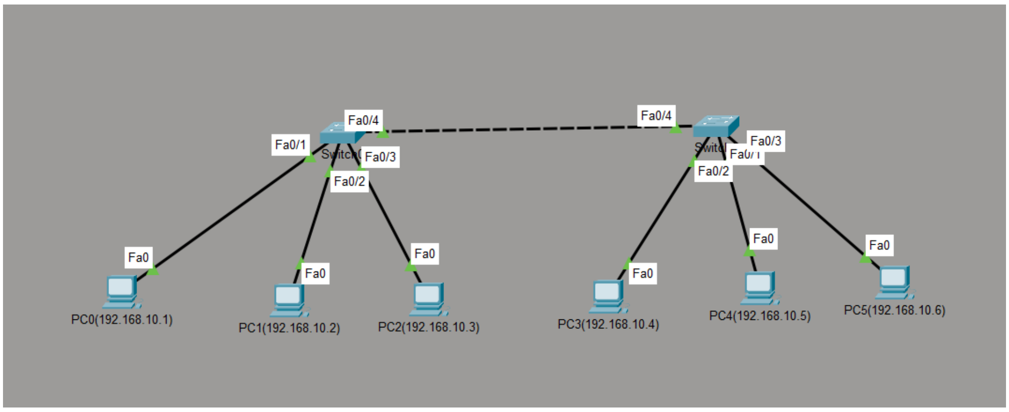
1. Objectives

- To study and practice the principles of VLAN.
- Create multiple VLANs and assign corresponding ports.
- Create a trunk line between 2 switches for paired VLANs.
- By implementing the **port security** understands Layer 2 Security.
- Understand the implementation of **EtherChannel**.
- implement **Single-Arm Routing** for communication between different IP subnets.

2. Preparation

- 2 switches (Like 2960).
- 6 PC.
- Each switch is connected to 3 PCs.
- Connect the 2 switches.

3. Description of laboratory experiment set-up



4. Experiment 1 – Basic Configuration

a. Create VLANs and name them

Repeat the following commands on both switches (Switch0 and Switch1):

Switch> **enable**

Switch# **configure terminal**

Switch(config)# **vlan 10**

Switch(config-vlan)# **name EEE**

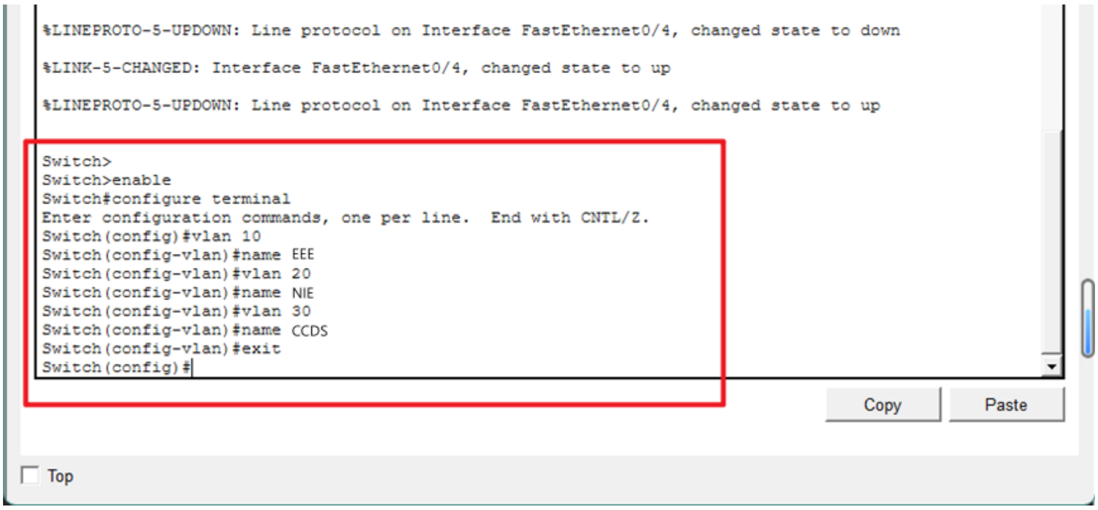
Switch(config-vlan)# **vlan 20**

Switch(config-vlan)# **name NIE**

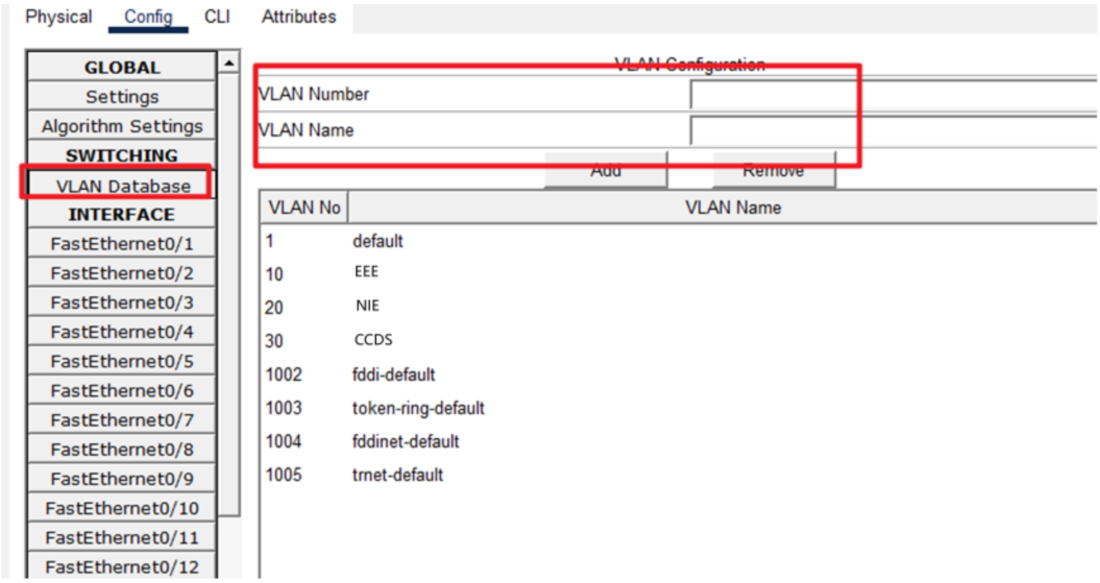
Switch(config-vlan)# **vlan 30**

Switch(config-vlan)# **name CCDS**

Switch(config-vlan)# **exit**



Or you can choose to change the config on both switches (Switch0 and Switch1):



b. Assign access port

Execute on Switch0 and Switch1 respectively:

i. Assign PC0 / PC3 to VLAN 10 (interface Fa0/1):

Switch(config)# **interface FastEthernet0/1**

Switch(config-if)# **switchport mode access**

Switch(config-if)# **switchport access vlan 10**

ii. Assign PC1 / PC4 to VLAN 20 (interface Fa0/2):

Switch(config)# **interface FastEthernet0/2**

Switch(config-if)# **switchport mode access**

Switch(config-if)# **switchport access vlan 20**

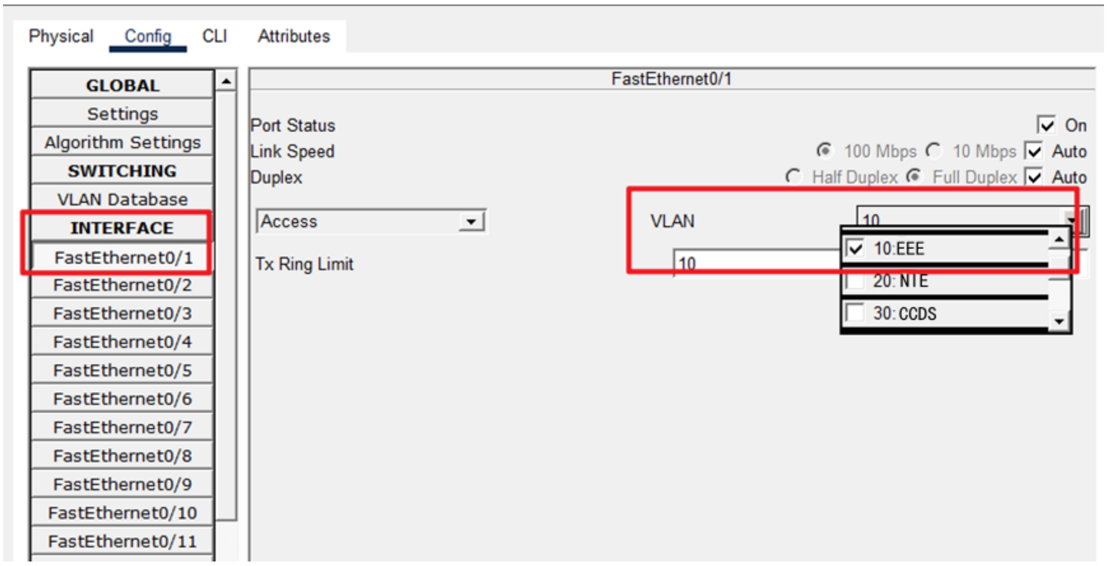
iii. Assign PC2 / PC5 to VLAN 30 (interface Fa0/3):

Switch(config)# **interface FastEthernet0/3**

Switch(config-if)# **switchport mode access**

Switch(config-if)# **switchport access vlan 30**

Or change according FastEthernet config:



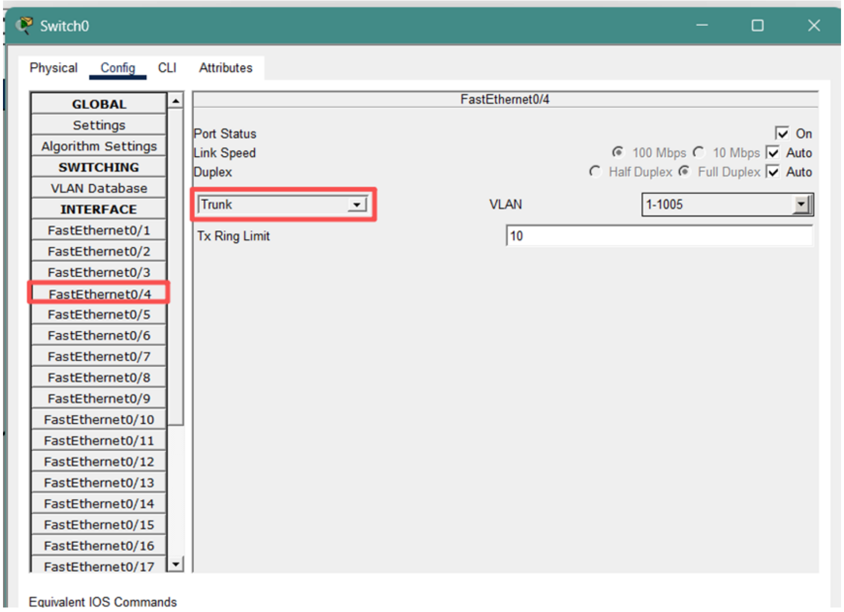
c. Configure trunk

Execute on Switch0 and Switch1 respectively:

Switch(config)# **interface FastEthernet0/4**

Switch(config-if)# **switchport mode trunk**

Or change according ethernet config:



Note: After executing this command, the link lights in Packet Tracer may temporarily turn red and then green, indicating that the Spanning Tree Protocol (STP) is converging again.

d. **View VLAN & trunk & interfaces information**

Enter "enable" at the "Switch> " (At Switch's CLI interface),

i. To view VLAN allocation, input: "**show vlan brief**":

ii. Check whether Trunk is enabled: "**show interfaces trunk**"

iii. Check interface status: "**show ip interface brief**"

e. **Practice & Thinking Questions**

i. Test 1 (VLAN Cross-Switch): Please ping the VLAN 10 host of Switch 1 from the VLAN 10 host of Switch 0. What is the result? What does this indicate about the role of the Trunk link?

The result is successful pinging. This indicates that the Trunk link can carry and transmit data traffic from multiple different VLANs over a single physical link, achieving logical interconnection within the same department (same VLAN) across physical switches.

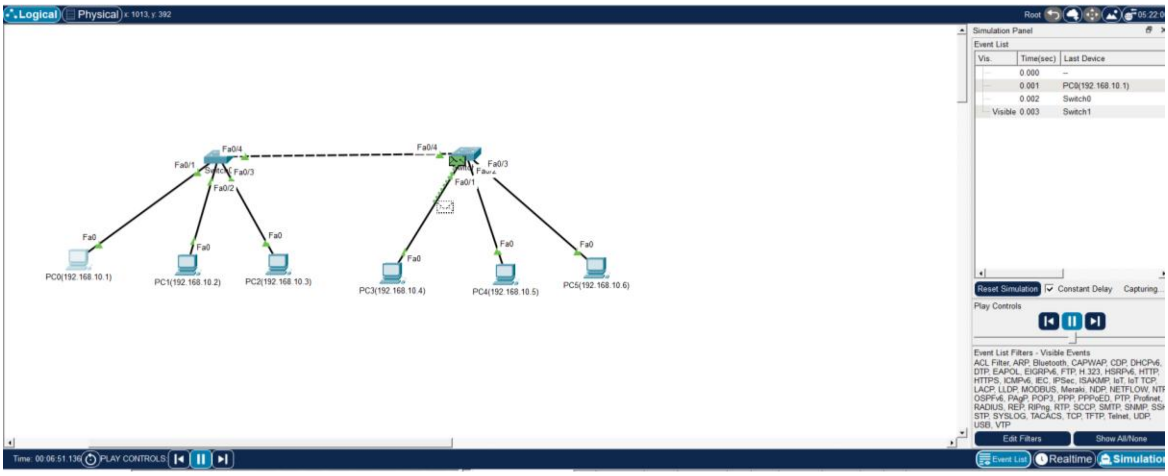
ii. Test 2 (Communication across Different VLANs): Please ping the host on VLAN 20 of Switch 0 from the host on VLAN 10 of Switch 0. What is the result? Explain why this result occurs, taking into account the MAC address table (show mac address-table).

The result is that pinging is not possible (Request Timeout). By using the command "show mac address-table", it can be observed that even if two hosts are connected to the same switch, their MAC addresses are bound to different VLAN IDs (such as VLAN 10 and VLAN 20). Switches are layer 2 devices that divide independent broadcast domains based on VLANs. Without the intervention of layer 3 routing devices, switches strictly isolate traffic from different VLANs when performing table lookups. Therefore, data frames from VLAN 10 cannot cross over to VLAN 20.

If you want to prove that "only VLAN 10 can receive", you can use the Simulation mode of Packet Tracer:

Click the clock icon in the lower right corner to switch to Simulation mode.

In the "Edit Filters" section, only check the ARP option.



Let PC0 ping PC3.

Click the "Play" button, and you will see the green packet (ARP) flying out like an envelope.

Observation: You will notice that the envelope only flies to ports within the same VLAN, while a red "X" appears above other VLAN ports, indicating that they have been blocked.

iii. Test 3: Change one end of the Trunk link between two switches back to Access mode, observe the changes in network communication, and consider what consequences would arise from a mismatch in negotiation between the two ends.

Phenomenon: All inter-departmental communications across switches have been interrupted (communications within the same department on the same switch are not affected).

Principle explanation: This can lead to severe protocol mismatch issues. The data frames sent by the Access port are untagged, while the Trunk port expects to receive data frames with 802.1Q tags (except for the Native VLAN). This confusion not only prevents VLAN data across switches from being correctly identified and forwarded, but also often triggers CDP (Cisco Discovery Protocol) errors (indicating Native VLAN Mismatch) on Cisco switches, and may even trigger the protection mechanism of the Spanning Tree Protocol, resulting in ports being placed in Err-disable state and ultimately causing the entire backbone link to fail.

iv. Q1(Broadcast Domain Isolation): When no VLAN is configured, if a PC sends an ARP broadcast request, which devices in the network can receive it? After configuring VLANs 10, 20, and 30, if a PC within VLAN 10 sends an ARP broadcast, who else can receive it? Please briefly describe how VLAN improves network performance and security.

When no VLAN is configured: all devices in the network (the entire network) can receive the broadcast.

After configuring VLAN: Only devices within VLAN 10 (including VLAN 10 hosts across switches) can receive it.

Performance improvement: VLAN narrows the broadcast domain, reducing the consumption of bandwidth and host CPU by useless broadcast messages across the entire network, effectively preventing broadcast storms.

Security enhancement: Logical isolation between departments has been achieved, preventing packets from different VLANs from being directly sniffed, monitored, or accessed, thereby enhancing data confidentiality.

v. Q2(The Essence of Trunk): If we do not configure a Trunk interface, but instead use three network cables to connect the VLAN 10, 20, and 30 interfaces of two switches respectively, can we achieve the same communication effect? If so, why do we still use Trunk technology?

The same communication effect can be fully achieved. However, if Trunk is not used, adding each additional VLAN requires consuming one physical port from each of two switches and one physical network cable. In enterprise networks with dozens or even hundreds of VLANs, this approach has extremely poor scalability and is a serious waste of hardware resources. Trunk technology greatly saves port and cabling costs by inserting 802.1Q tags into data frames to "multiplex" and transmit multiple VLAN data over a single link.

vi. Q3(No routing state): In the experiment, different VLANs cannot communicate with each other. If we want VLAN 10 and VLAN 20 to communicate with each other, what type of device do we need to introduce into the network topology? (To pave the way for the next stage of the single-arm router/layer 3 switch experiment)

It is necessary to introduce network layer (Layer 3) devices, such as routers (Router) or Layer 3 switches (Layer 3 Switch). Because different VLANs logically represent different IP subnets, routing tables must be used for Layer 3 forwarding across network segments.

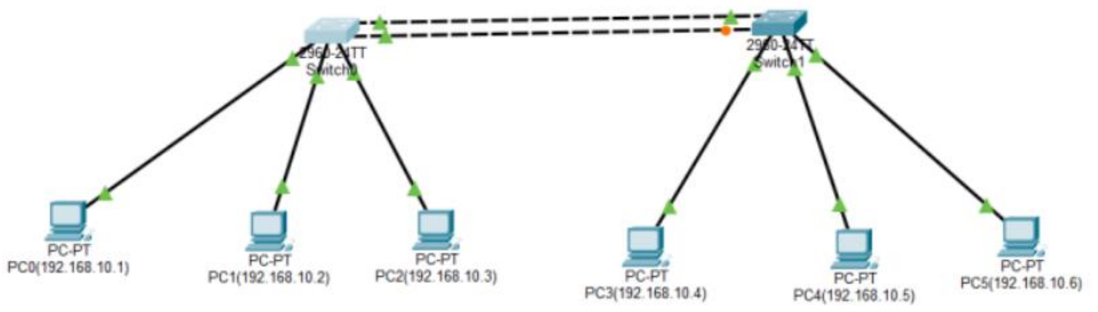
Note: Congratulations on completing Experiment 1 !

The following 3 experiments are based on this experiment (Experiment 1). Please save the .pkt file current as a template for subsequent experimental operations.

5. Experiment 2 -- EtherChannel and Spanning Tree Protocol (STP) Observation

a. Upgrade Topology and Observe the initial state of STP

Connect an additional crossover cable between two 2960 switches. After connecting the cable without further configuration, one port on one of the links will turn orange (Blocking state).



Enter *show spanning-tree* in the privileged mode to observe the root bridge election results and specific port statuses (blocked ports are displayed as BLK)

```
Switch>enable
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0090.2164.4740
             Cost        19
             Port        4(FastEthernet0/4)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00D0.58A4.7323
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/5            Altn BLK 19        128.5    P2p
Fa0/4            Root FWD 19        128.4    P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0090.2164.4740
             Cost        19
             Port        4(FastEthernet0/4)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
 --More--
```

Copy Paste

Top

b. Configure EtherChannel

i. In the global configuration mode of Switch 0 and Switch 1, enter the following identical configuration commands:

enable

show spanning-tree

1.Enter the two connected physical interfaces simultaneously.

interface range fastEthernet 0/4-5

2.Bundle these two ports into Channel Group 1 and set them to active negotiation mode.

channel-group 1 mode active

exit

Enter the newly created logical interface Port-Channel 1.

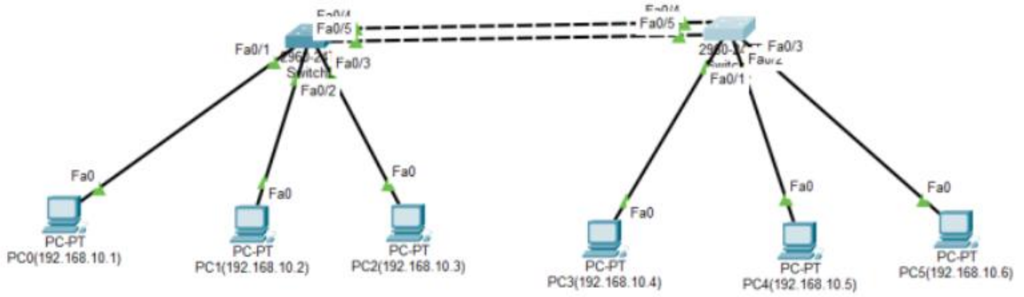
interface port-channel 1

#Configure this logical interface as Trunk mode to carry all VLANs.

switchport mode trunk

End

ii. Use the Fast Forward Time feature in Packet Tracer to manually accelerate the convergence of STP and LACP (Link Aggregation Control Protocol). Once the network converges, all four port indicators between the switches turn green. Communication between PC1 and PC4 remains normal.

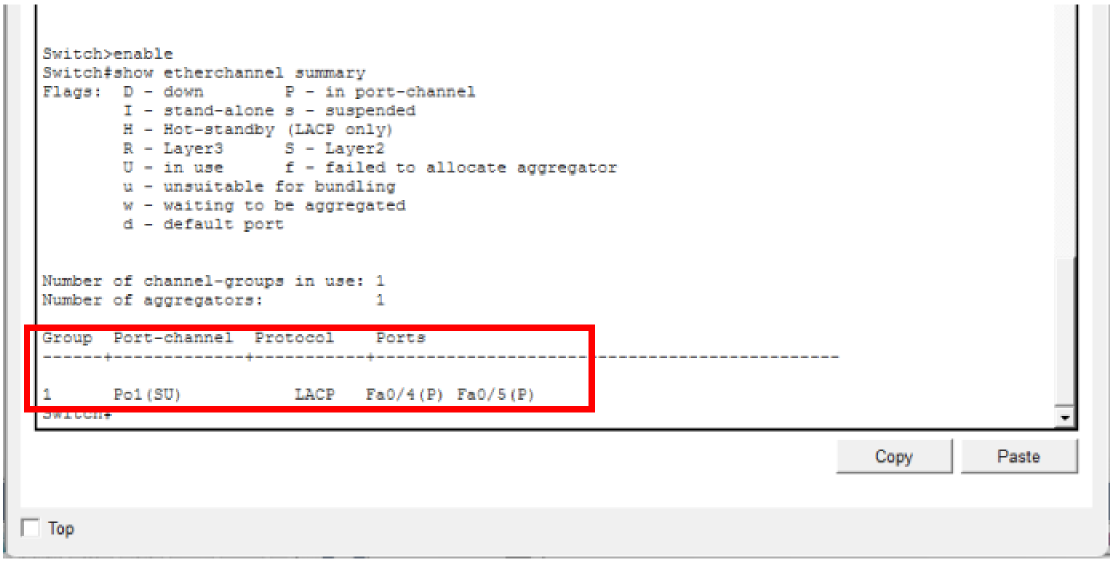


c. **Verification Procedures**

Execute the following three core commands in the privileged mode (Switch#) of Switch 0 or Switch 1:

i. Check Aggregation Summary Status

Command: Switch# **show etherchannel summary**

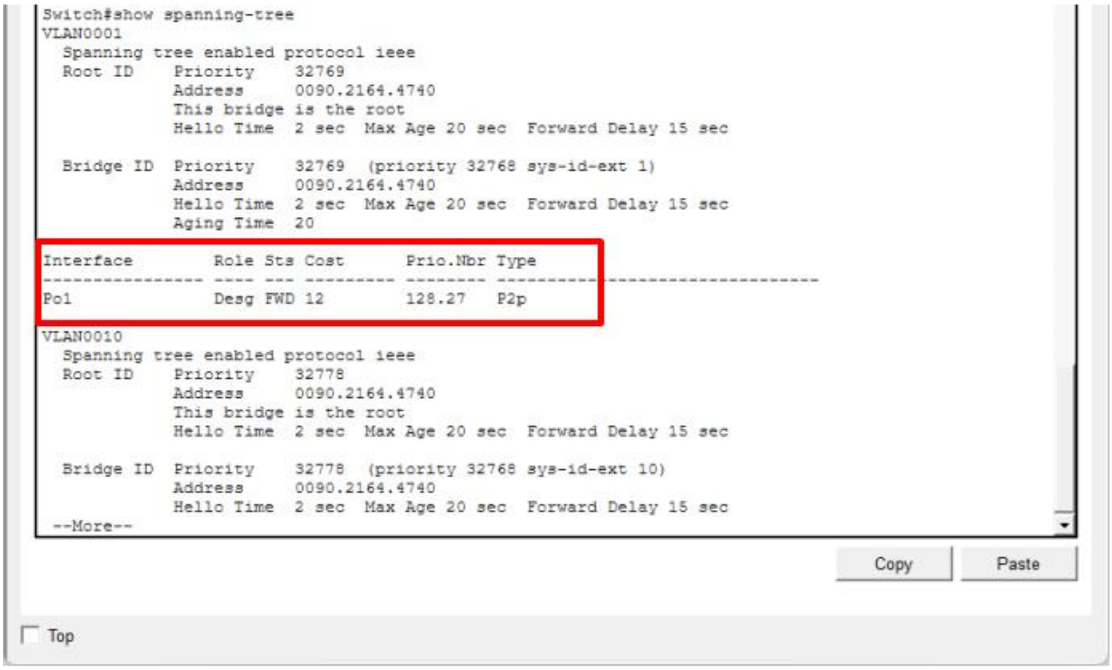


```
Switch>enable
Switch#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/4(P) Fa0/5(P)
Switch#
```


Check Group Status if Po1 is marked as (SU). S stands for Layer 2, and U stands for In Use, which signifies success.Check Ports Status if physical ports (e.g., Fa0/4, Fa0/5) are marked as (P), indicating they are successfully bundled into the Port-channel.

ii. Check Spanning Tree Protocol (STP) Status

command: Switch# **show spanning-tree**

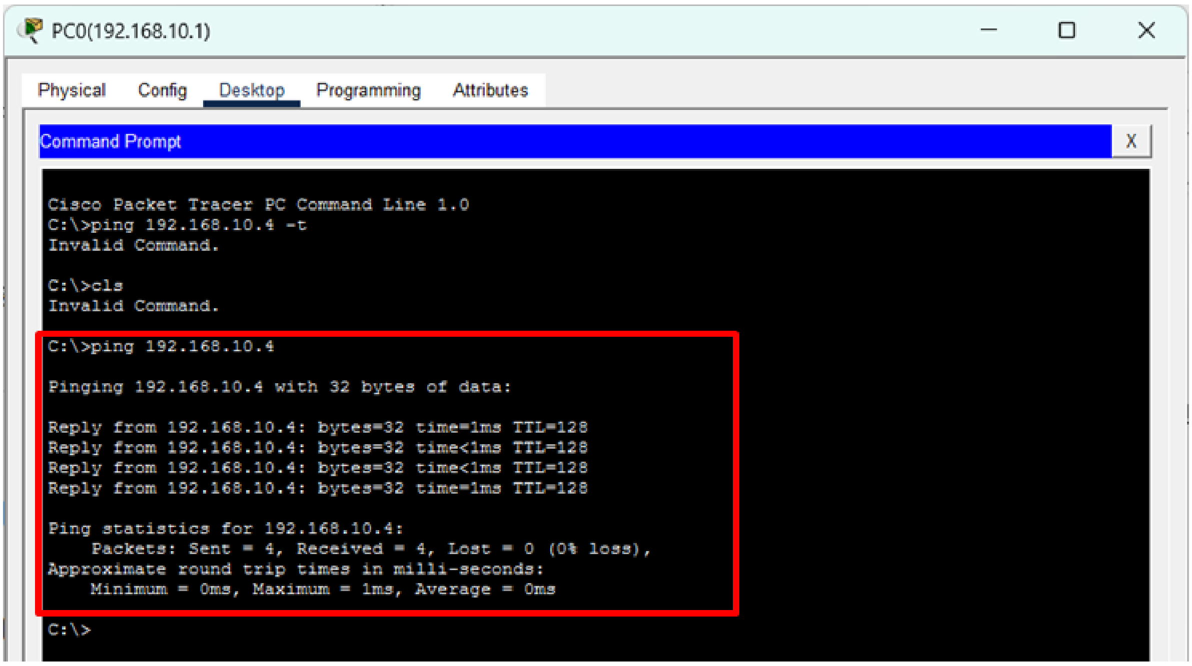


```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0090.2164.4740
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0090.2164.4740
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Po1              Desg FWD 12        128.27   P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0090.2164.4740
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0090.2164.4740
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
 --More--
```


The original physical ports Fa0/4 and Fa0/5 should disappear from the list, replaced by the logical interface Po1 in the FWD (Forwarding) state.

iii. Final Connectivity Test

Perform a continuous Ping test from PC0 (VLAN 10) to PC3 (VLAN 10). Ping responses are normal with 0% loss.



iv. Why did the orange port (blocked by STP) turn green after configuring EtherChannel?

STP treats the bundled physical lines as a single logical link. Since there is only one logical path, the loop is eliminated, and STP no longer blocks the port.

7. Experiment 3 — Port Security Configuration

a. Port security configuration command (CLI)

Please click on Switch0 to enter the CLI tab, and then enter the following commands in sequence:

Switch> **enable**

Switch# **configure terminal**

Switch(config)# **interface FastEthernet0/3**

\# The port must first be set to Access mode (Port Security does not support Trunk or dynamic ports)

Switch(config-if)# **switchport mode access**

Enable port security function

Switch(config-if)# **switchport port-security**

Limit the maximum number of MAC addresses allowed to be learned to 1

Switch(config-if)# **switchport port-security maximum 1**

Set to Sticky, permanently binding the currently connected PC's MAC address to the configuration

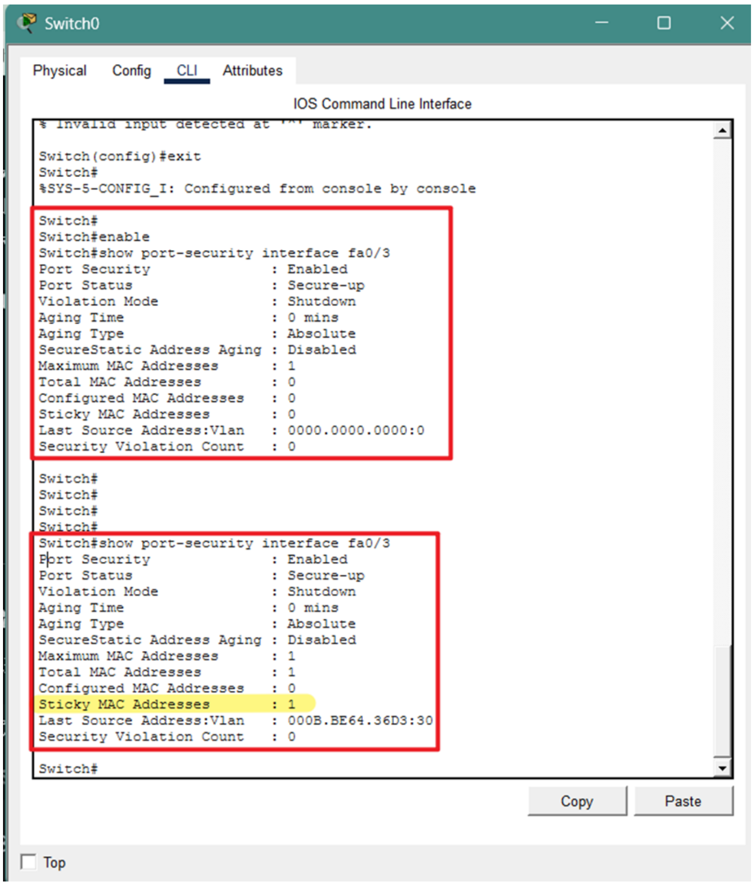
Switch(config-if)# **switchport port-security mac-address sticky**

Set the violation triggering action (default is shutdown, which means closing the port)

Switch(config-if)# **switchport port-security violation shutdown**

b. Ensure that the switch "captures" the identity of legitimate hosts

Ping other PCs in the net. Making sure that the switch remember the MAC address. You can confirm that the Sticky MAC Addresses has changed to 1 by entering "show port-security interface fa0/3" on the switch.

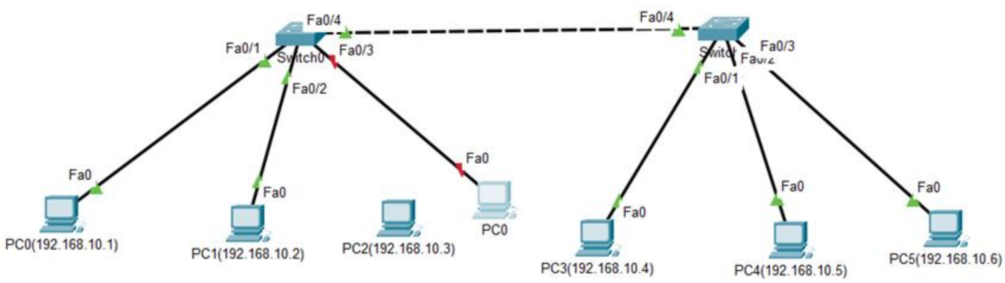


c. Simulating "hacker" access

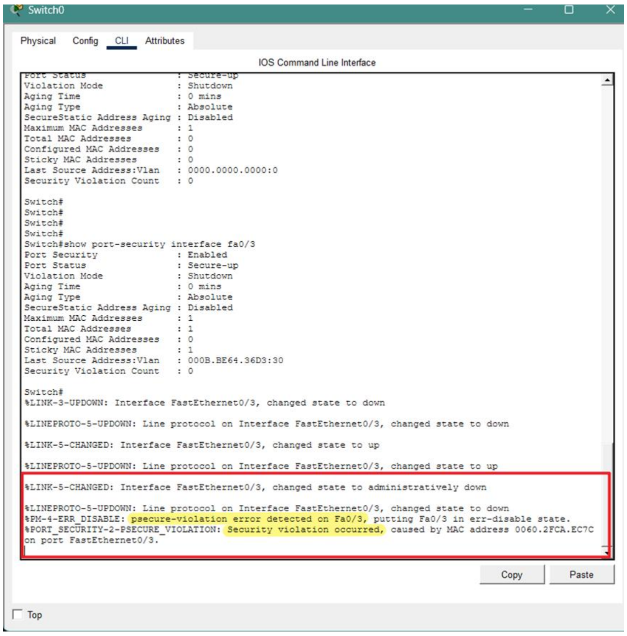
Disconnect: Use the delete tool (shortcut key Del) to remove the connection between Switch0 and the original PC2.

Connect a new device: Add a new laptop or PC and connect it to Fa0/3 of Switch0.

Trigger: When configuring the IP address for the hacker's PC, you will notice that the light on the Fa0/3 interface instantly changes from green to red.



At this time, you can view the following information by checking the CLI interface of the switch:



d. Phenomenon observation and recovery

When a port is disabled due to violations, it enters a special state known as "err-disabled".

i. View violation records

Enter the follow commands at Switch's CLI tab(Switch#):

Switch# show port-security interface FastEthernet0/3

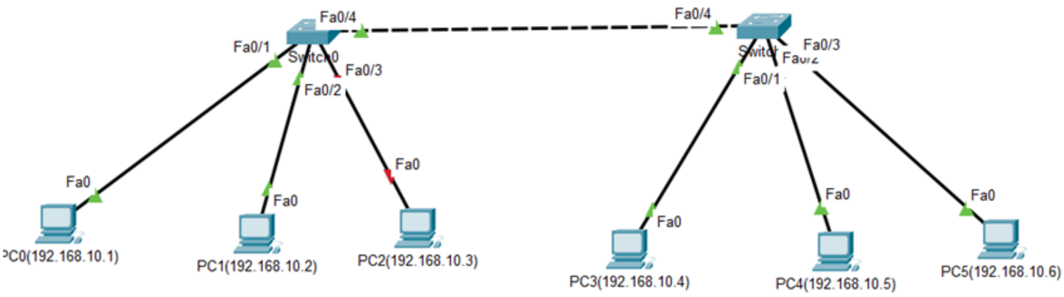
You will see in the output:

Port Status: Security-shutdown

Security Violation Count: 1

Last Source Address: [The MAC address of the hacker's PC will be displayed here]

ii. Restore interface



First, reconnect the originally legal PC.

Enter the following command at switch's CLI tab:

Switch# configure terminal

Switch(config)# interface FastEthernet0/3

Switch(config-if)# shutdown

Switch(config-if)# no shutdown

It can be observed that the original PC has restored normal connection.

Question: If you directly enter "not shutdown" instead of "shutdown" first, will the original PC be able to restore normal connection? What is the status of the interface? Why?

No. The interface status is err-disabled. When a port security violation occurs, the switch places the port into an err-disabled state to protect the network. In Cisco IOS, the no shutdown command is used to enable an administratively closed port, but it cannot clear a security-triggered error state. The shutdown command acts as a manual reset that clears the error flags. Without performing shutdown first, the switch logic keeps the security lock active, ignoring any no shutdown attempts.

e. Auto Recovery

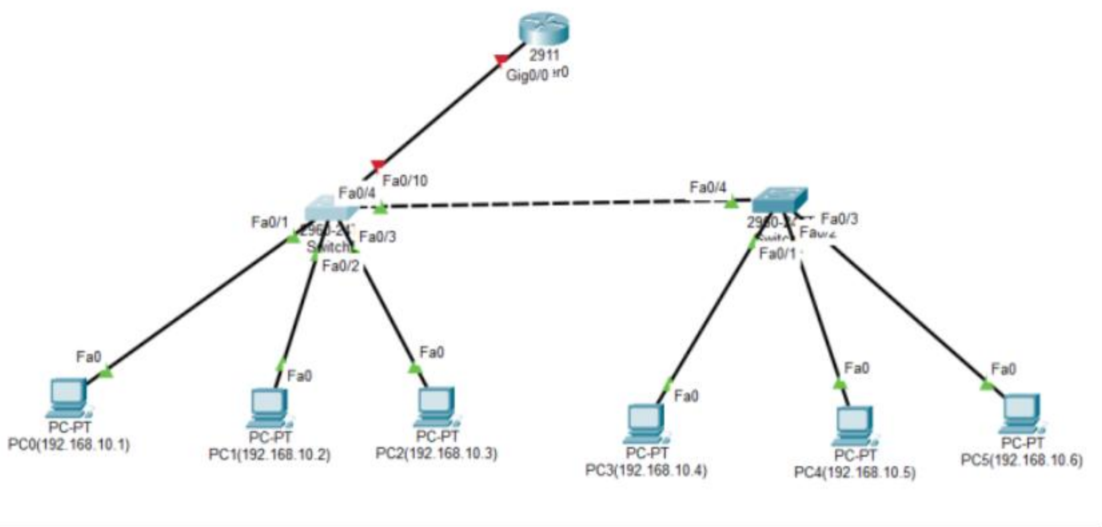
In real-world production environments, manually closing and reopening hundreds of ports can be extremely cumbersome. Cisco provides an automatic recovery function:

```
Switch(config)# errdisable recovery cause psecure-violation
# Set the waiting time for automatic recovery (e.g. 30 seconds)
Switch(config)# errdisable recovery interval 30
```

8. Experiment 4 —— Single-Arm Routing Configuration

a. Upgrade Topology

Use a straight-through cable to connect Switch0 (Fa0/10) to Router 2911. (GigabitEthernet 0/0). Initially, both ends of the cable will appear red; this is normal because router interfaces are disabled by default.



b. Configure Single-Arm Routing

i. Switch Side: Backbone Link Configuration

To allow the link connecting to the router to carry tagged packets for multiple VLANs, the port must be set to Trunk mode. Enter the CLI for Switch0:

Switch>enable

#User Mode → Privileged Mode: Gain authorization to view and modify configurations.

Switch# configure terminal

#Privileged Mode → Global Configuration Mode: Prepare to modify switch parameters.

Switch(config)# interface fastEthernet 0/10

#Interface Configuration Mode: Specify the Fa0/10 port connected to the router.

Switch(config-if)# switchport mode trunk

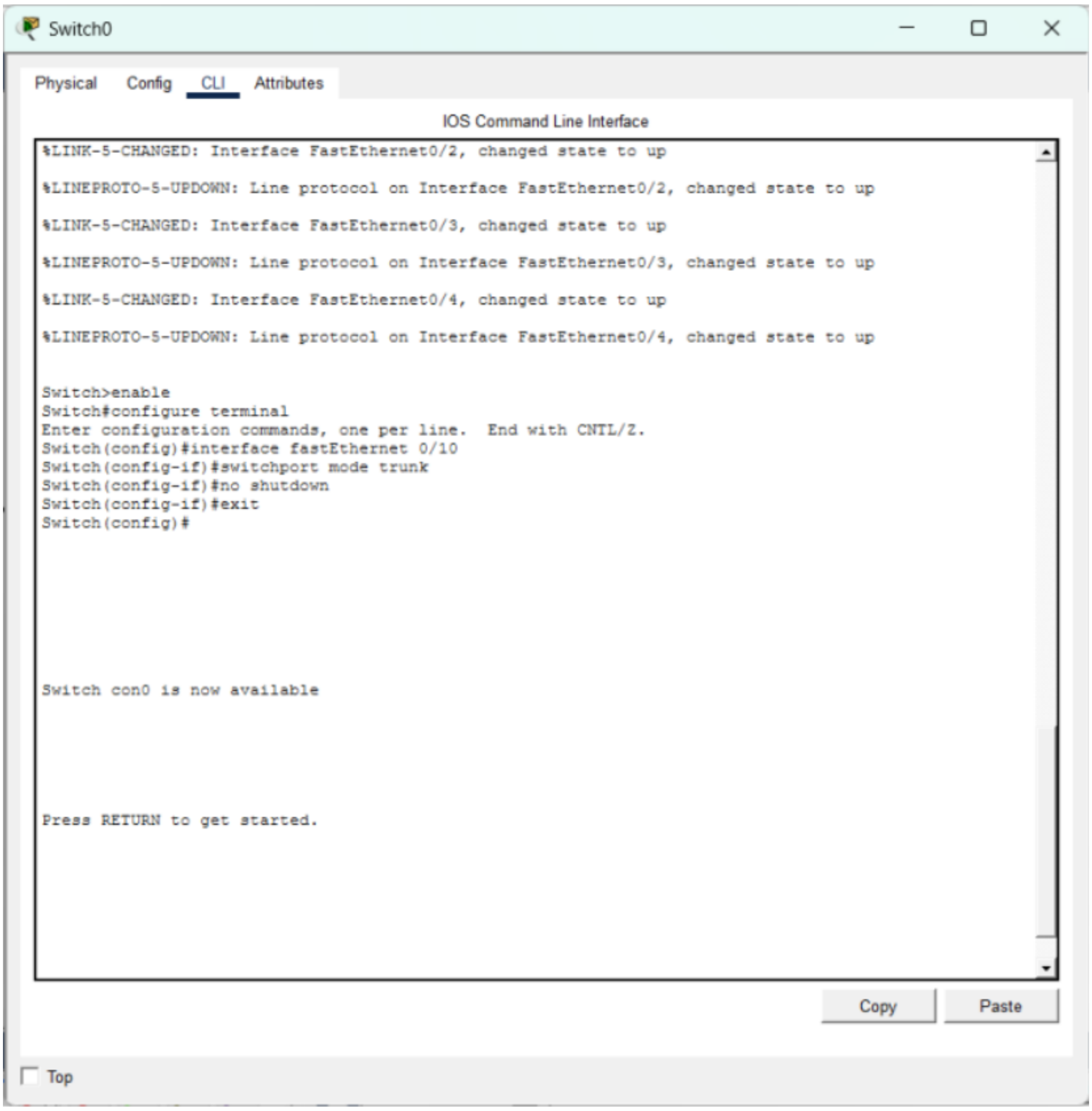
#Configure Trunking: Allow the link to carry data for multiple VLANs (10 and 20).

Switch(config-if)# no shutdown

#Activate Interface: Ensure the physical port is turned on.

Switch(config-if)# exit

#Exit Interface: Return to the previous Global Configuration level.



ii. Router Side: Sub-interface and Gateway Configuration

The router provides an independent Layer 3 entry point for each VLAN by dividing logical sub-interfaces.Enter the CLI for the Router:

Router> enable

Enter privileged mode: Ready to start configuring the router.

Router# configure terminal

Enter global configuration mode: Ready to enter configuration commands.

Router(config)# interface gigabitEthernet 0/0,

Physical interface mode: Enter the main physical interface to perform the enable operation.

Router(config-if)# no shutdown

Enable the physical master switch: The connection lines in the topology diagram will change from red to green at this point.

Router(config-if)# exit

Return to global configuration mode: Ready to create subinterfaces.

--- Subinterface 10 ---

Router(config)# interface gigabitEthernet 0/0.10

#Enter subinterface mode: Create a logical interface dedicated to VLAN 10.

Router(config-subif)# encapsulation dot1Q 10

Encapsulation protocol: Bind the 802.1Q protocol and specify that it belongs to VLAN 10.

Router(config-subif)# ip address 192.168.10.254 255.255.255.0

#Configure Gateway IP: Provide a routing exit for VLAN 10 hosts

Router(config-subif)# exit

#Return: Prepare to configure the next VLAN.

--- Sub-interface 20 ---

Router(config)# interface gigabitEthernet 0/0.20,

#Sub-interface Mode: Create a logical interface dedicated to VLAN 20.

Router(config-subif)# encapsulation dot1Q 20,

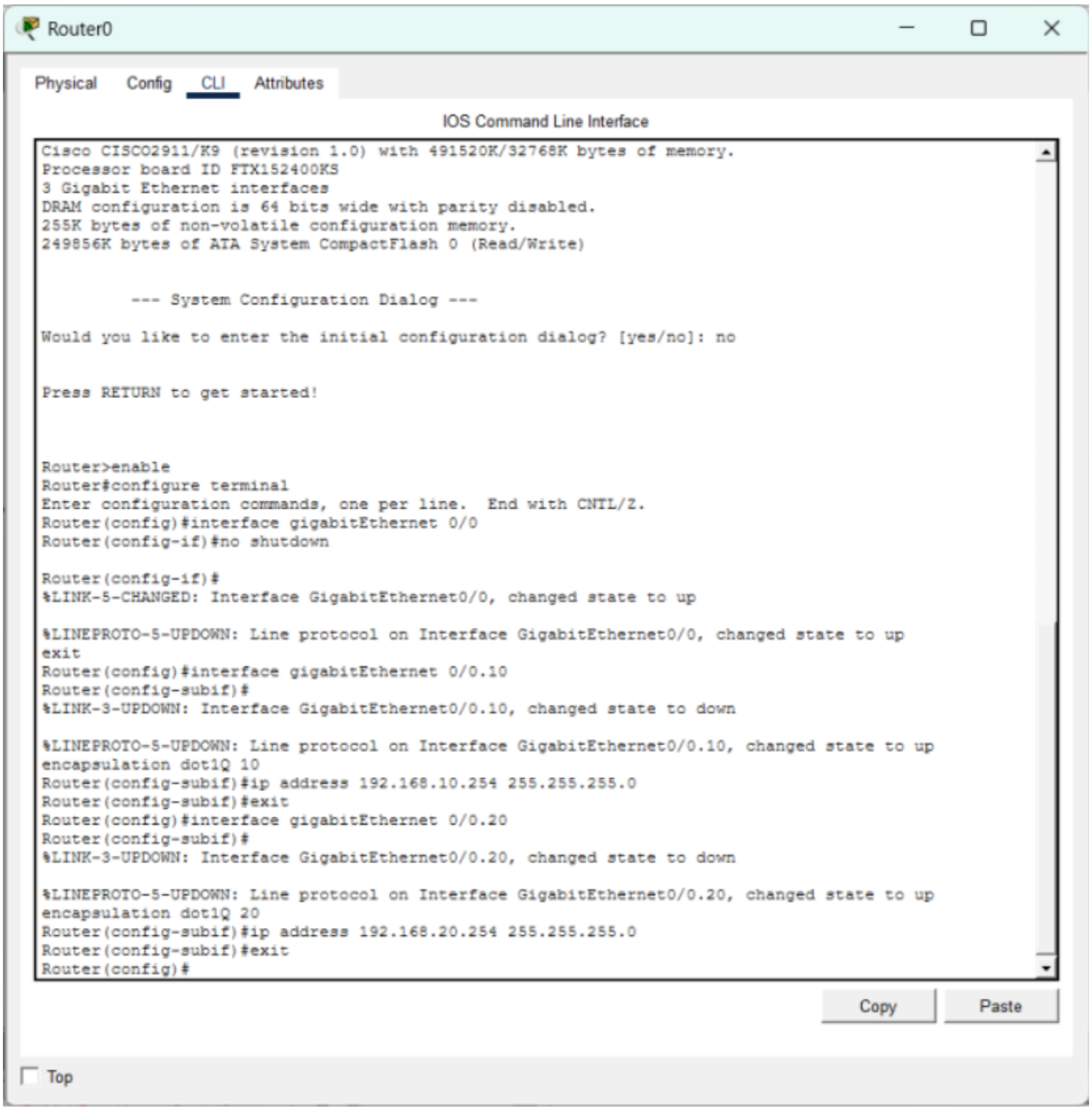
#Encapsulation Protocol: Bind the 802.1Q protocol to VLAN ID 20.

Router(config-subif)# ip address 192.168.20.254 255.255.255.0,

#Configure Gateway IP: Provide a routing exit for VLAN 20 hosts.

Router(config-subif)# end

#Finish Configuration: Exit back directly to Privileged Mode.



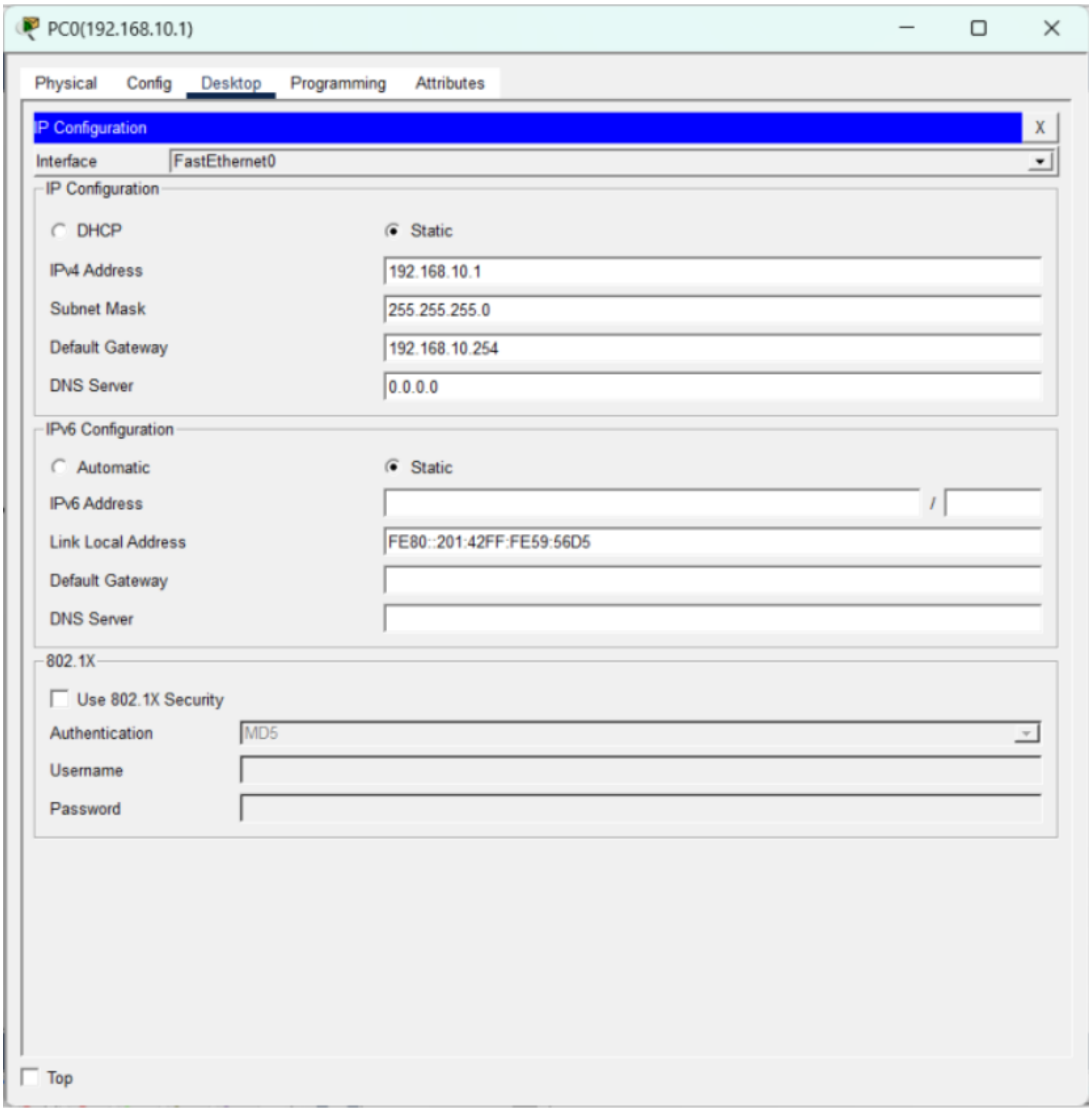
c. Verification Procedures

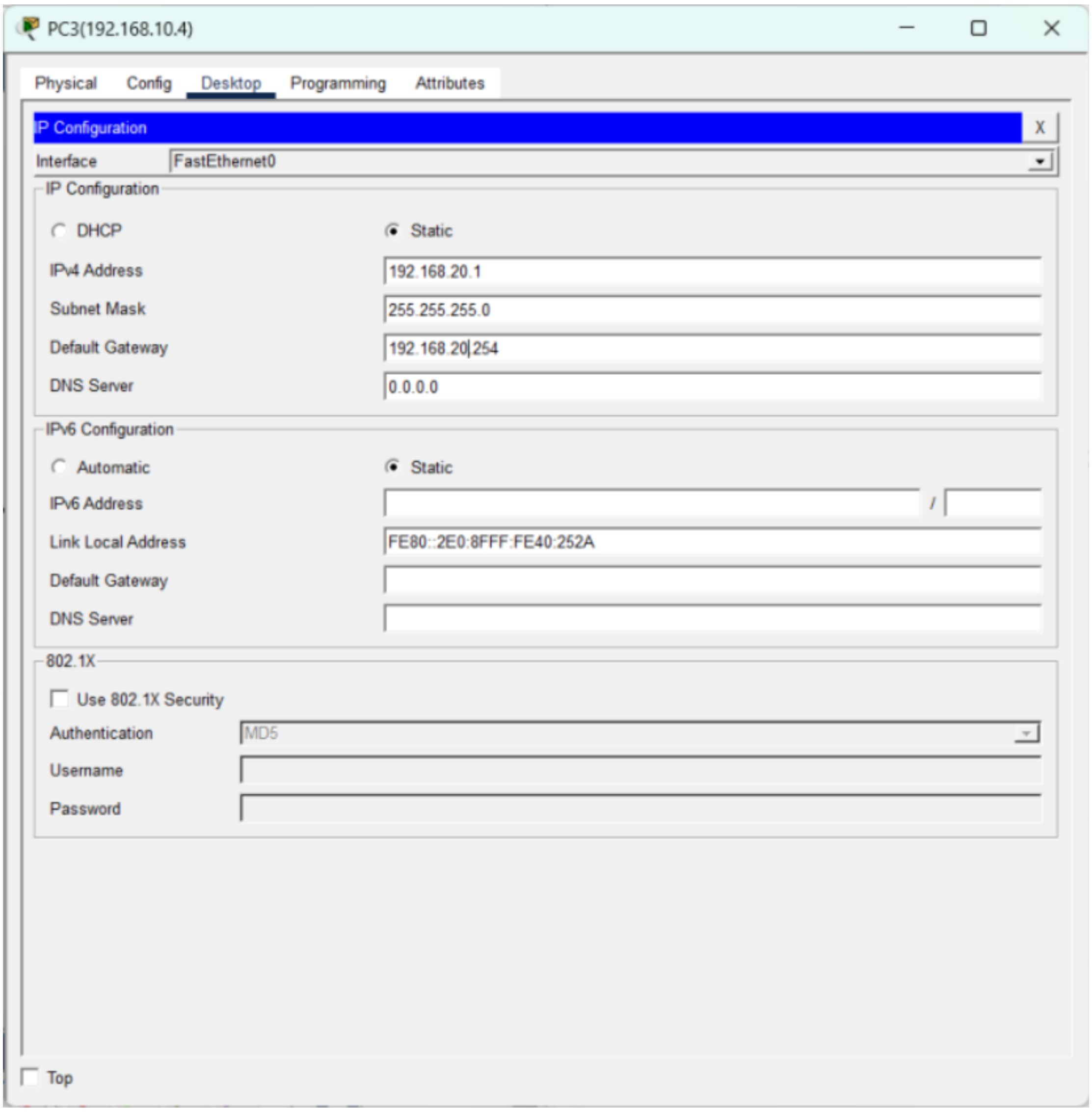
i. Terminal Configuration Verification

Click the PC icon, select the Desktop tab, and click IP Configuration to enter details:

PC0 (VLAN 10): IP `192.168.10.1`, Default Gateway must point to `192.168.10.254`.

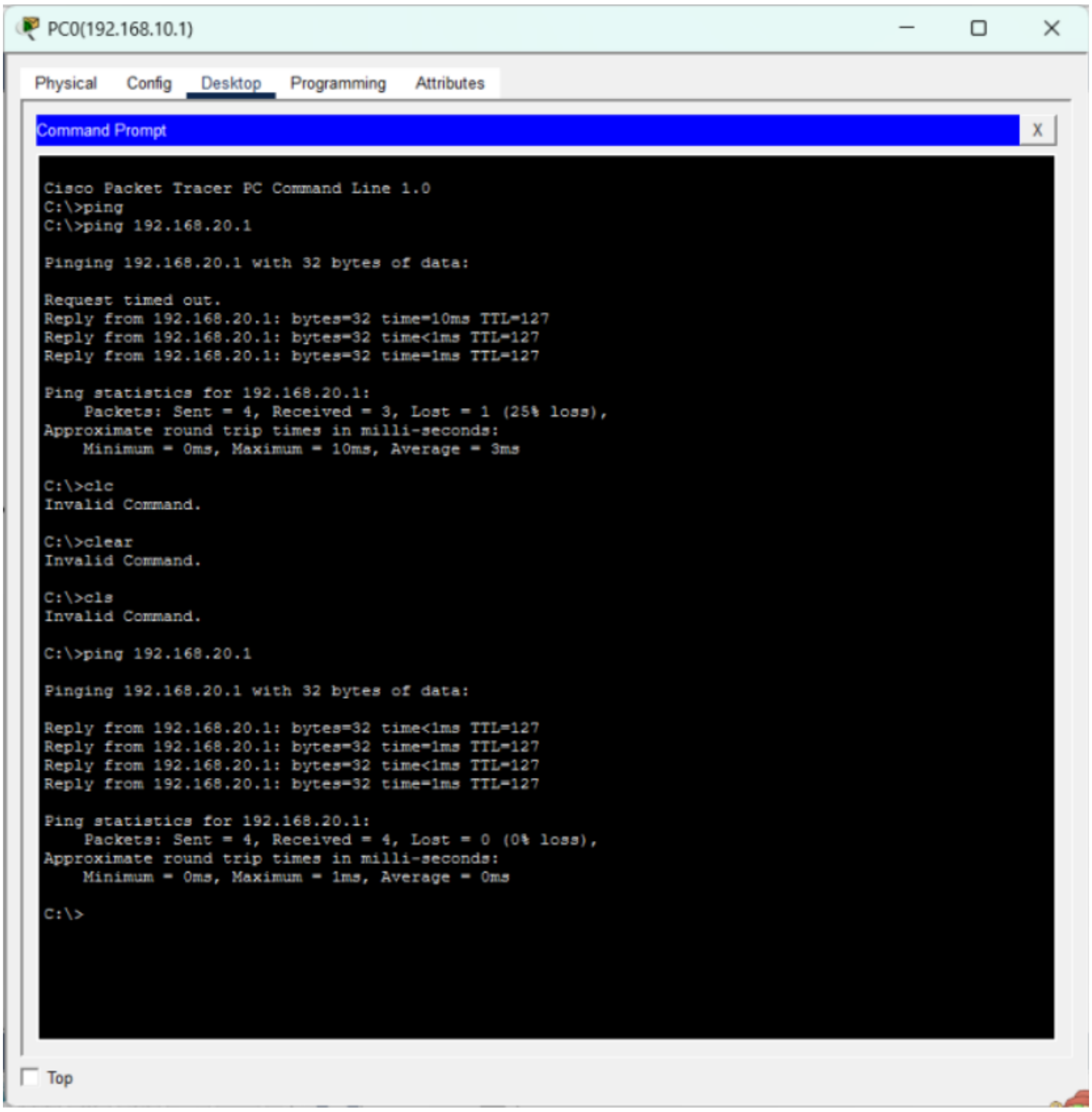
PC3 (VLAN 20): IP `192.168.20.1`, Default Gateway must point to `192.168.20.254`.


PC0(192.168.10.1)
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.1
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:42FF:FE59:56D5
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top



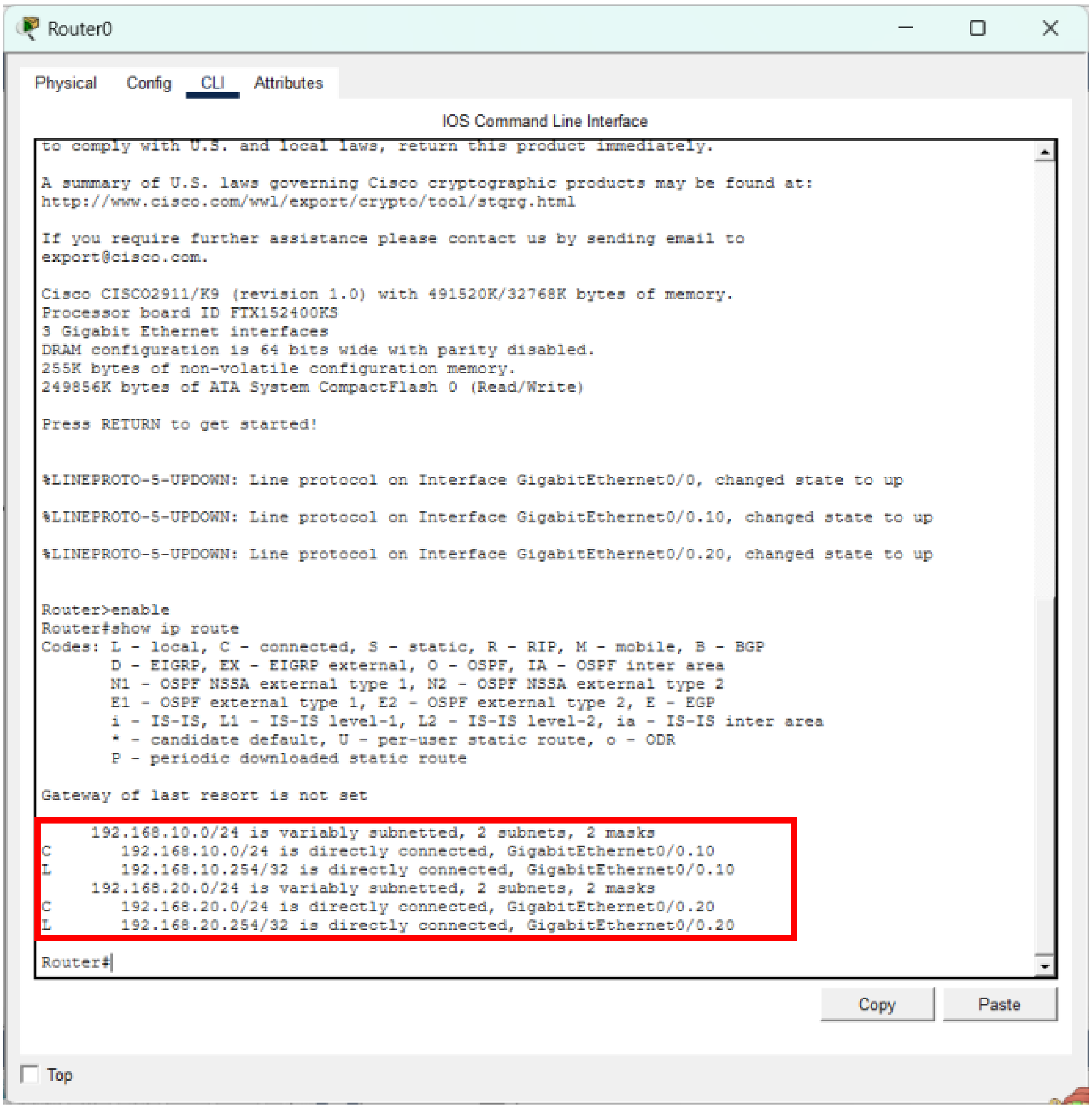
ii. Connectivity Test (Ping Test)

Open the Command Prompt on PC0 and enter ping 192.168.20.1. Due to ARP requests, the first 1-2 packets may "Timed out," followed by "Reply from...". Successful replies prove cross-segment communication is successful.



iii. Routing Table Verification

Enter show ip route on the router CLI. You should see two routes marked with C (Connected) pointing to Gi0/0.10 and Gi0/0.20 respectively.



iv. Why is Trunk mode mandatory for the Switch-Router link?

Access ports can only carry traffic for a single VLAN. Since the router needs to receive and send traffic for both VLAN 10 and VLAN 20 over a single cable, the link must be a Trunk to distinguish between different VLAN tags (802.1Q).

v. What happens if the Default Gateway on the PC is configured incorrectly?

The PC will be able to ping hosts within its own VLAN, but it will fail to reach other network segments. Without the correct gateway (the router's sub-interface IP), the PC does not know where to send packets destined for external networks.

vi. What are the limitations of the Single-Arm Routing approach in a high-traffic environment?

Since all Inter-VLAN traffic must pass through a single physical cable, this link can become a "bottleneck." Additionally, it represents a single point of failure.

Part 4. Network Address Translation (NAT) Basics and Configuration (additional part)

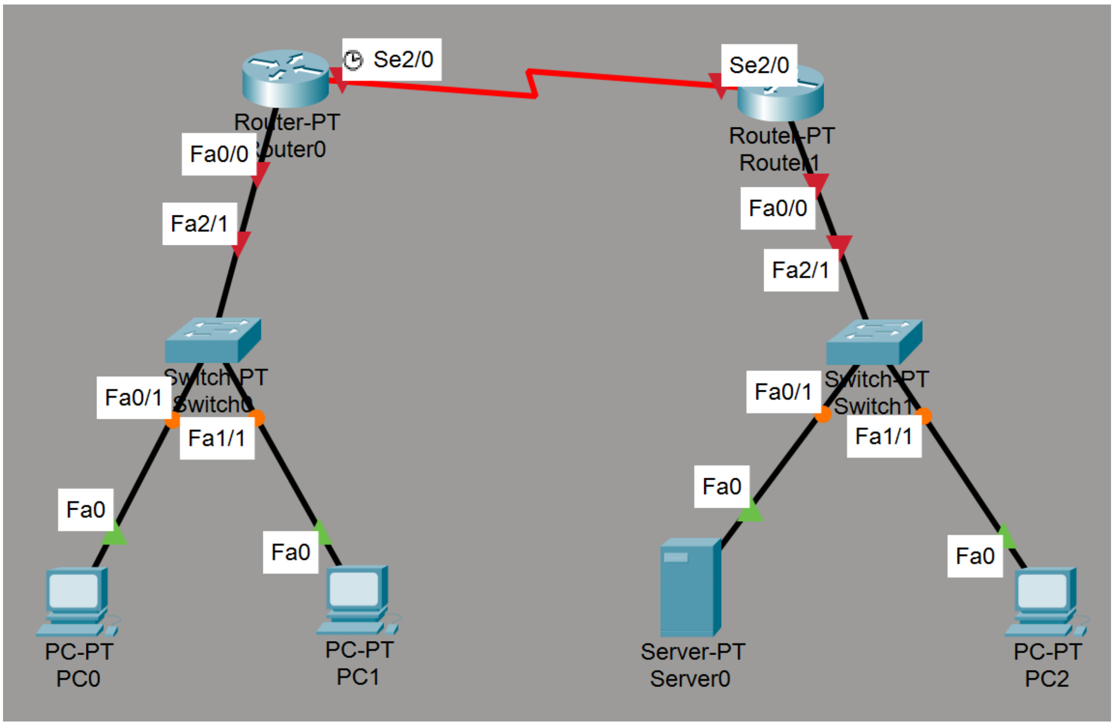
1. Objectives

- Learn and understand the working principles of Network Address Translation (NAT), distinguishing between Private Networks and Public Networks.
- Master basic router configurations, including the setup of Serial communication and Static Routes.
- Configure Static NAT on the router to achieve one-to-one mapping from private IPs to public IPs.
- Verify NAT protection features and analyze the network communication process through packet tracing and the NAT Table.

2. Preparation

- Cisco Packet Tracer software
- 2 Routers (Model: Router-PT, providing serial modules)
- 2 Switches (Model: Switch-PT)
- 3 PCs (simulating public and private network terminals)
- 1 Server (simulating an external Web server, providing HTTP services)

(For the convenience of the experiment, the department has been simplified to a single host.)

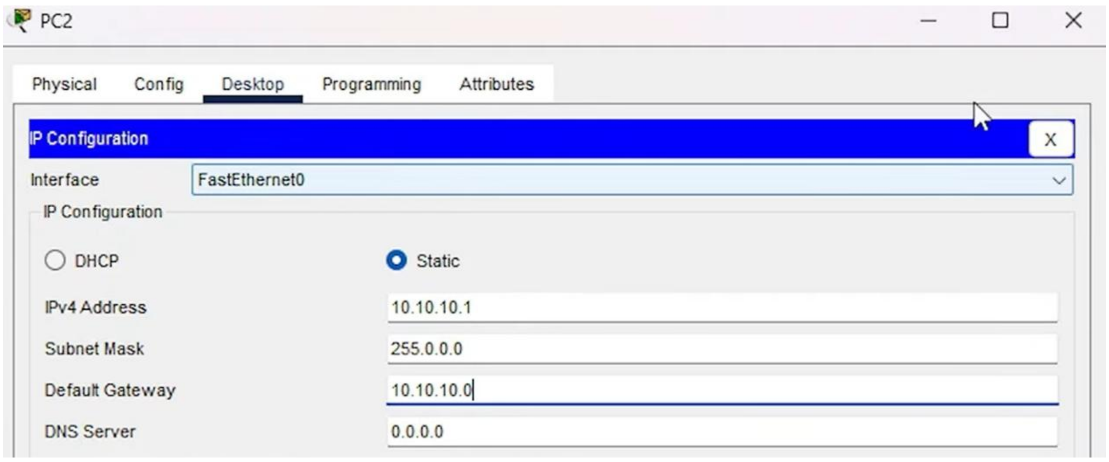


3. Experiment 1 Network Topology Setup and IP Address Allocation

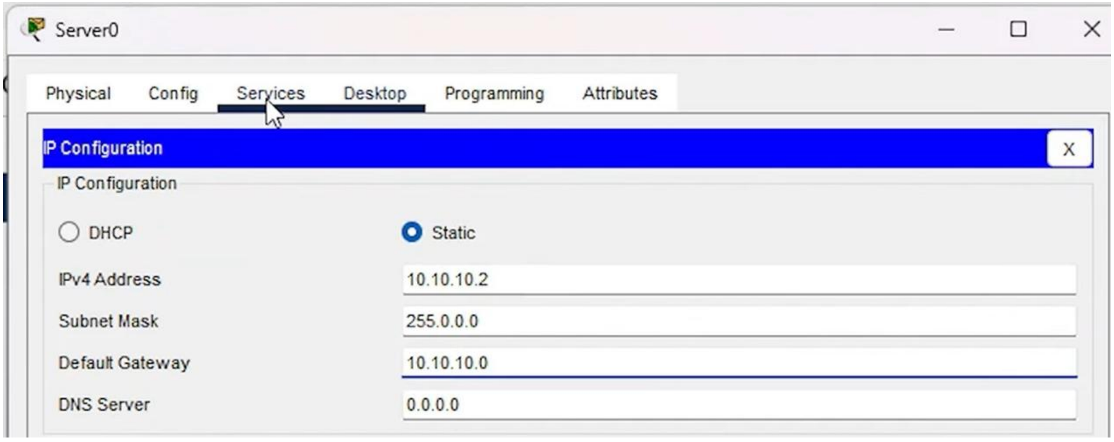
In this experiment, we will build two isolated network environments: a private network and a public network, and allocate IP addresses to the devices.

a. Private Network IP Address Allocation

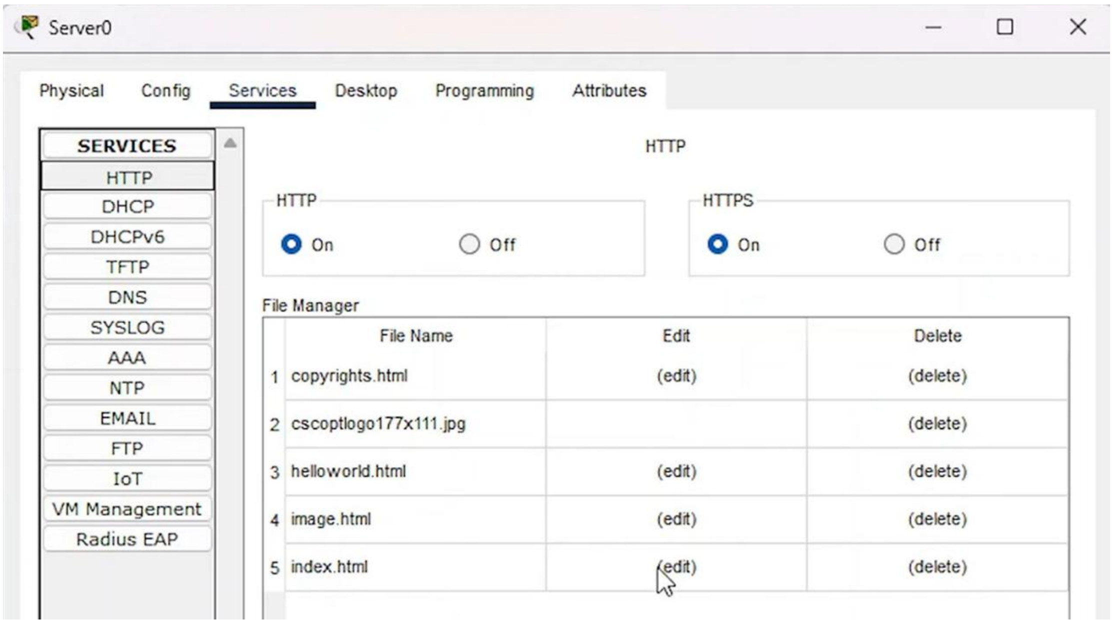
i. PC2: Open Desktop > IP Configuration. Set the IPv4 address to 10.10.10.1, subnet mask to 255.0.0.0, and default gateway to 10.10.10.0.

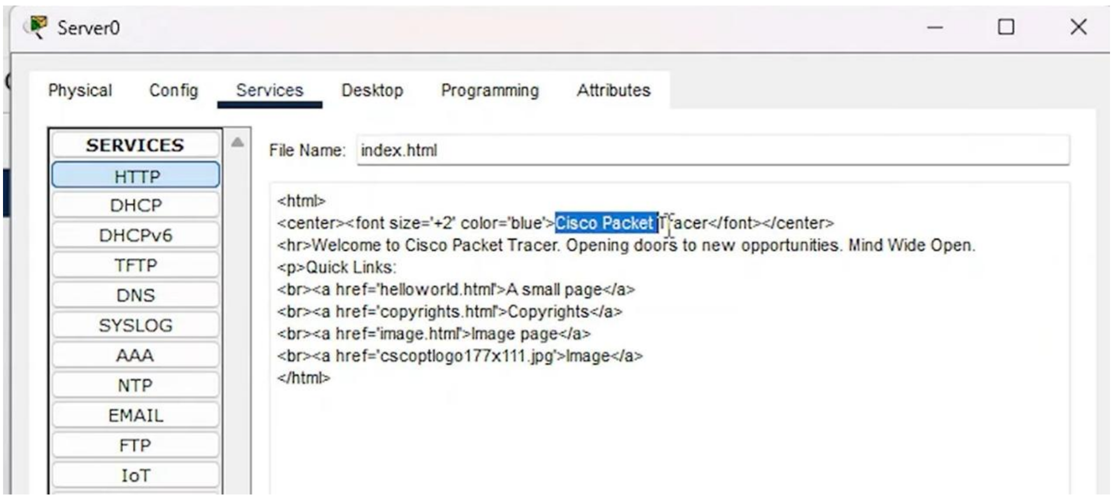


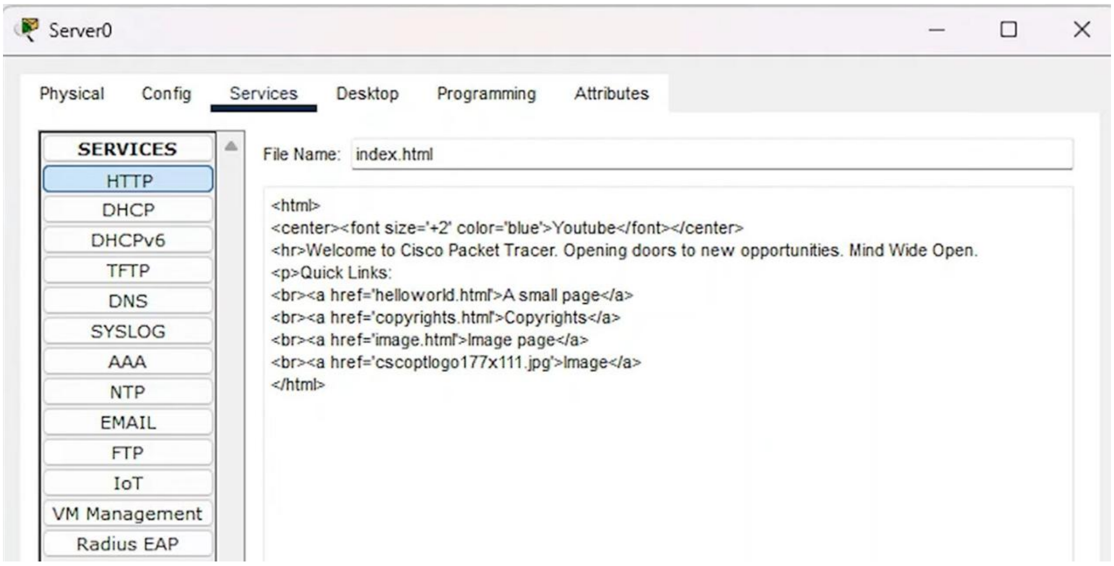
ii. Server (Server0): Open Desktop > IP Configuration. Set the IPv4 address to 10.10.10.2, subnet mask to 255.0.0.0, and default gateway to 10.10.10.0.



iii. Configure Web Service: Go to the Services tab of the server and ensure HTTP and HTTPS are turned on. Edit the index.html file, change its title text to "Youtube", and save it for subsequent testing.

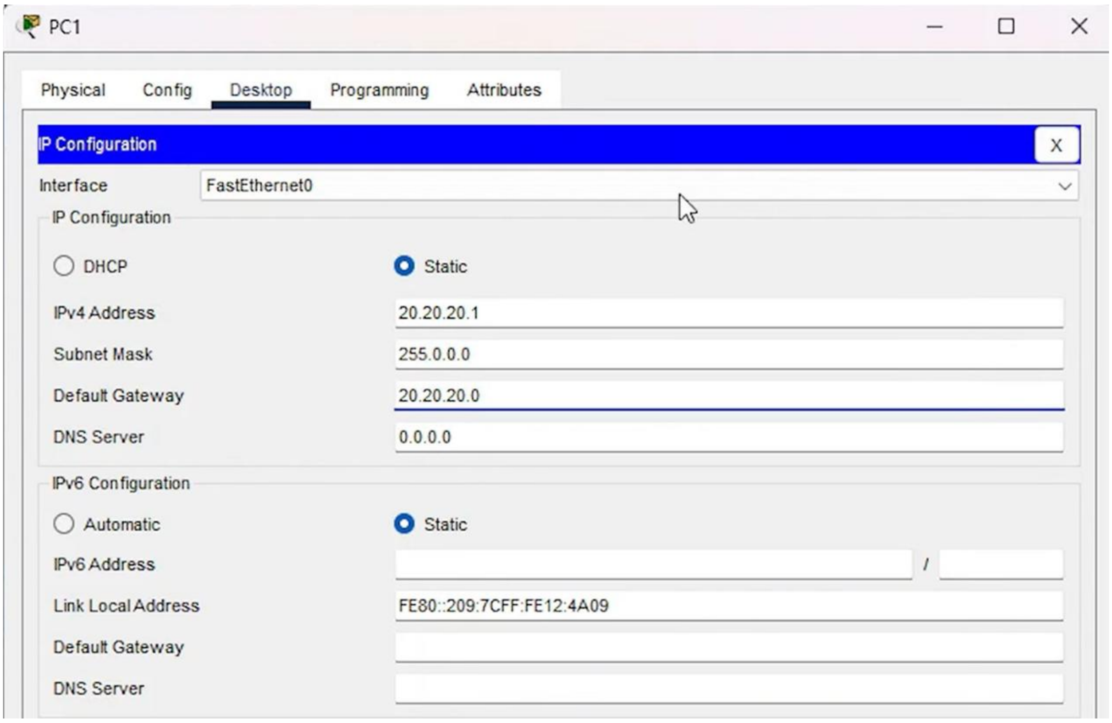




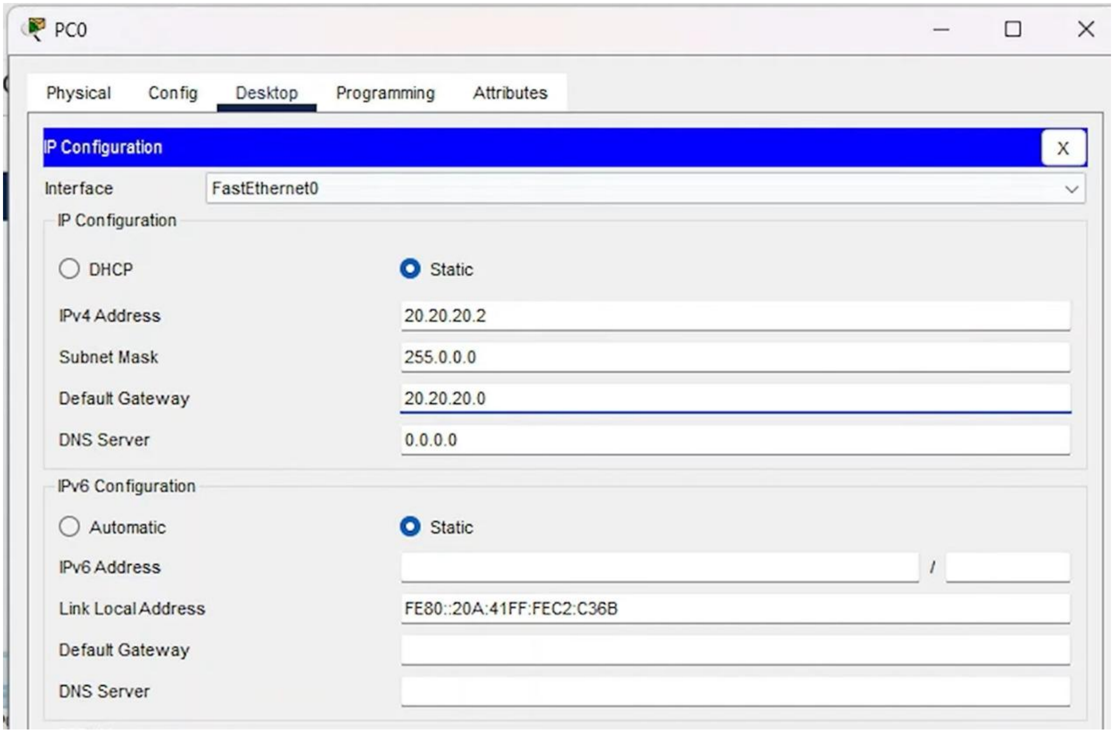


b. Public Network IP Address Allocation

i. PC1: Open Desktop > IP Configuration. Set the IPv4 address to 20.20.20.1, subnet mask to 255.0.0.0, and default gateway to 20.20.20.0.



ii. PC0: Set the IPv4 address to 20.20.20.2, subnet mask to 255.0.0.0, and default gateway to 20.20.20.0.



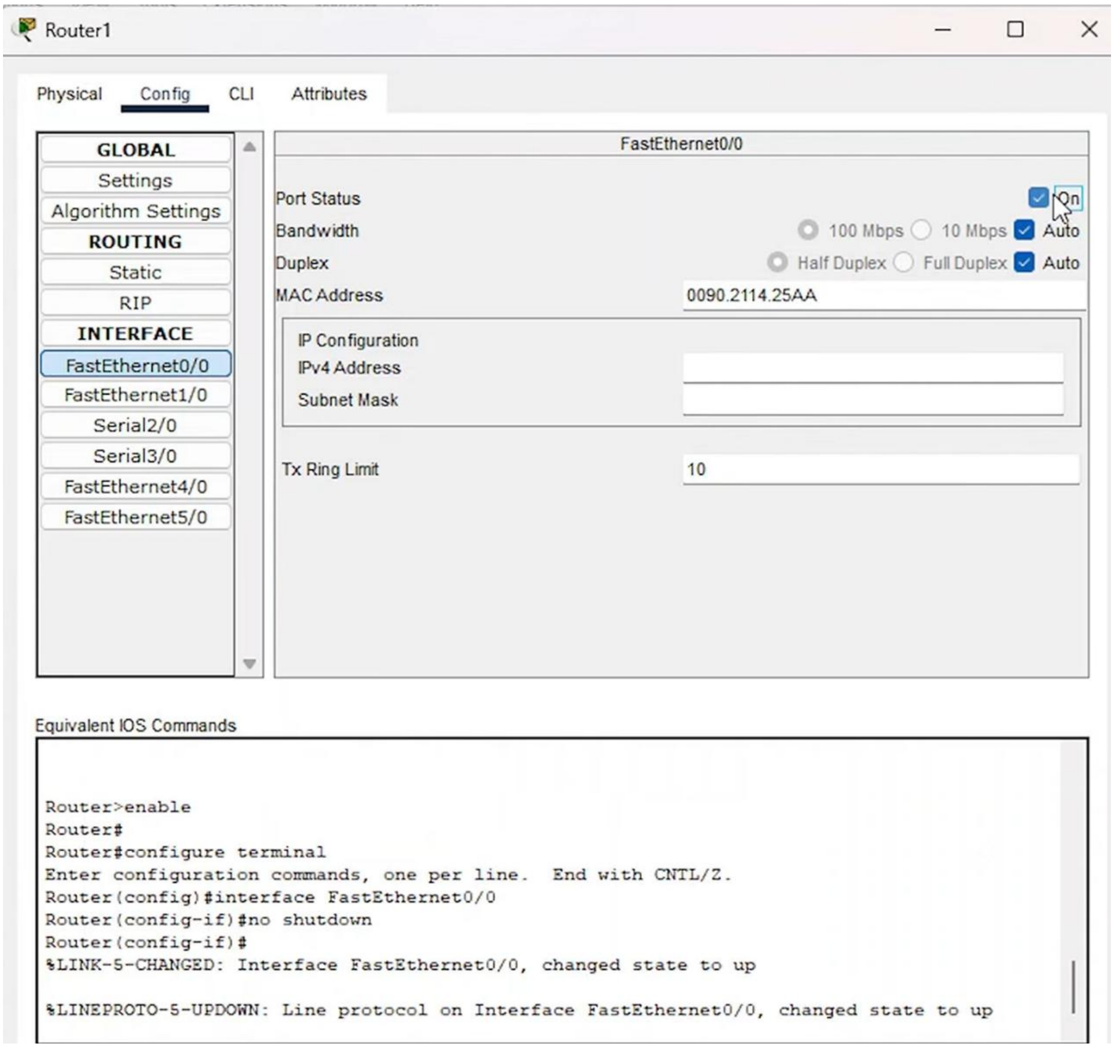
4. Experiment 2 Router Interface and Static NAT Configuration

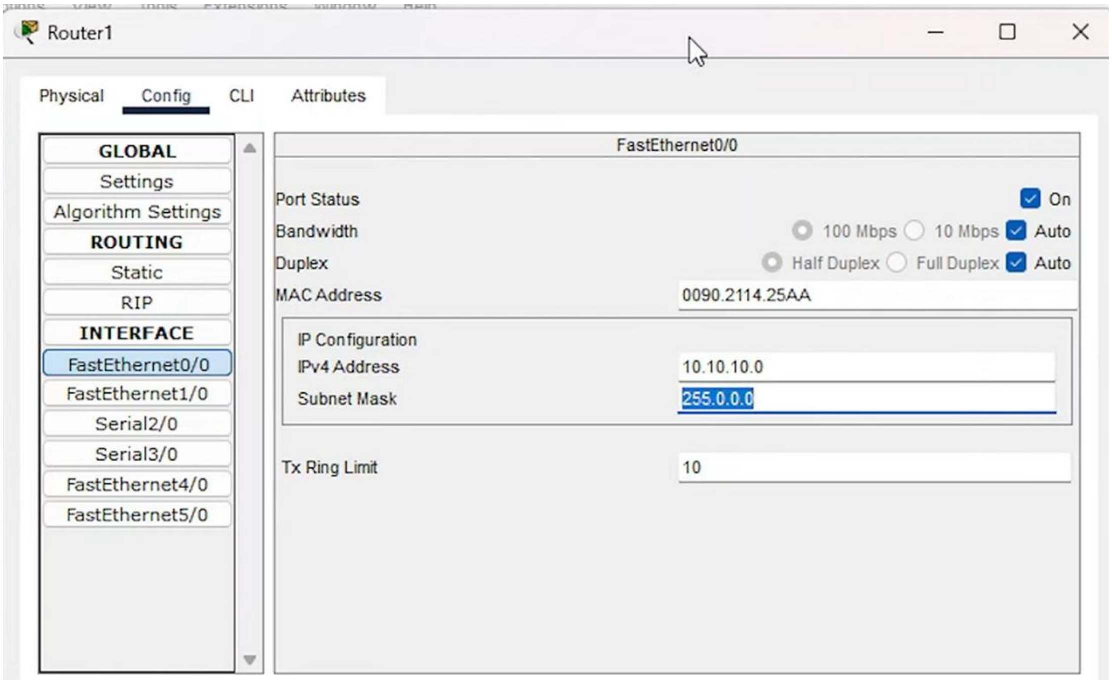
This session will establish the physical links of the routers and configure static NAT translation rules on the boundary router.

a. Basic Router Configuration

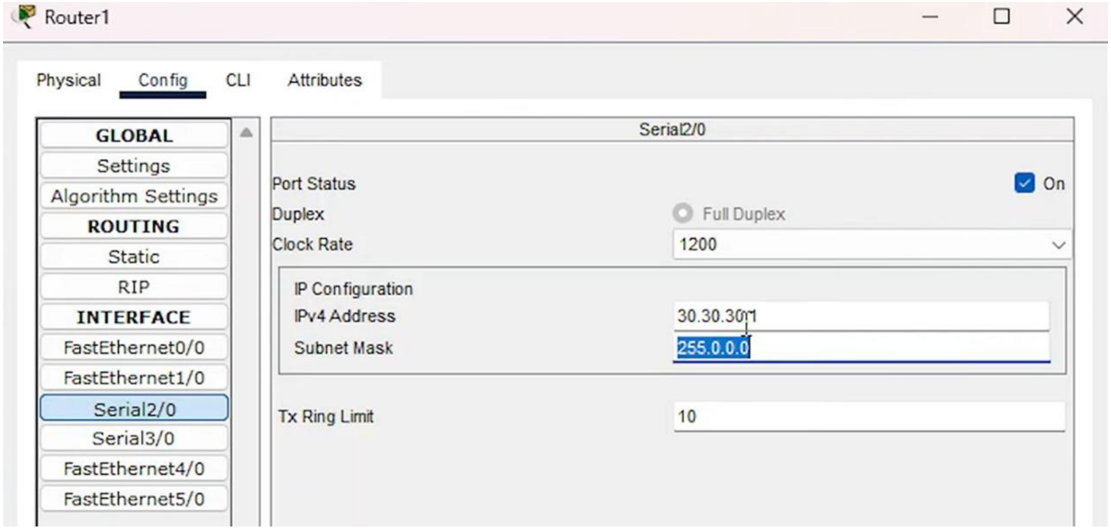
i. Router 1 (Private Network Side):

Go to the Config tab > Interface f0/0 (connected to the switch). Turn on Port Status and assign the IPv4 address 10.10.10.0 (which is the default gateway).


Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.2114.25AA
IP Configuration
IPv4 Address
Subnet Mask
Tx Ring Limit
10
Equivalent IOS Commands
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

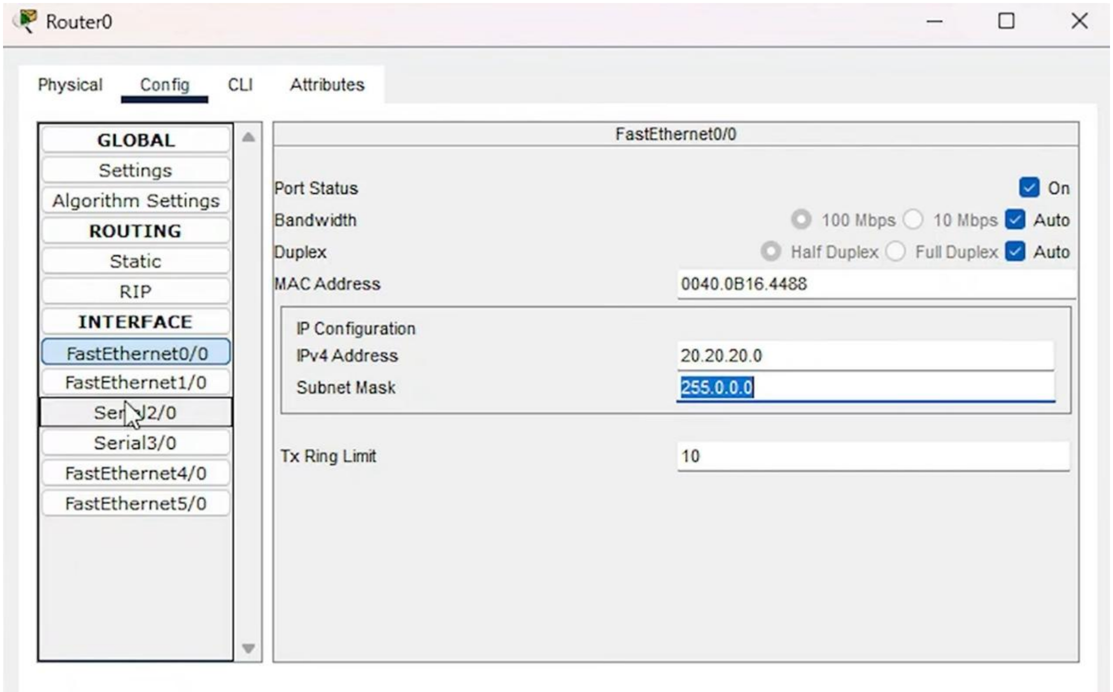

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.2114.25AA
IP Configuration
IPv4 Address
10.10.10.0
Subnet Mask
255.0.0.0
Tx Ring Limit
10

Go to Interface serial 2/0 (connected to Router 0). Turn on Port Status, assign the IPv4 address 30.30.30.1, and let it automatically generate the subnet mask.

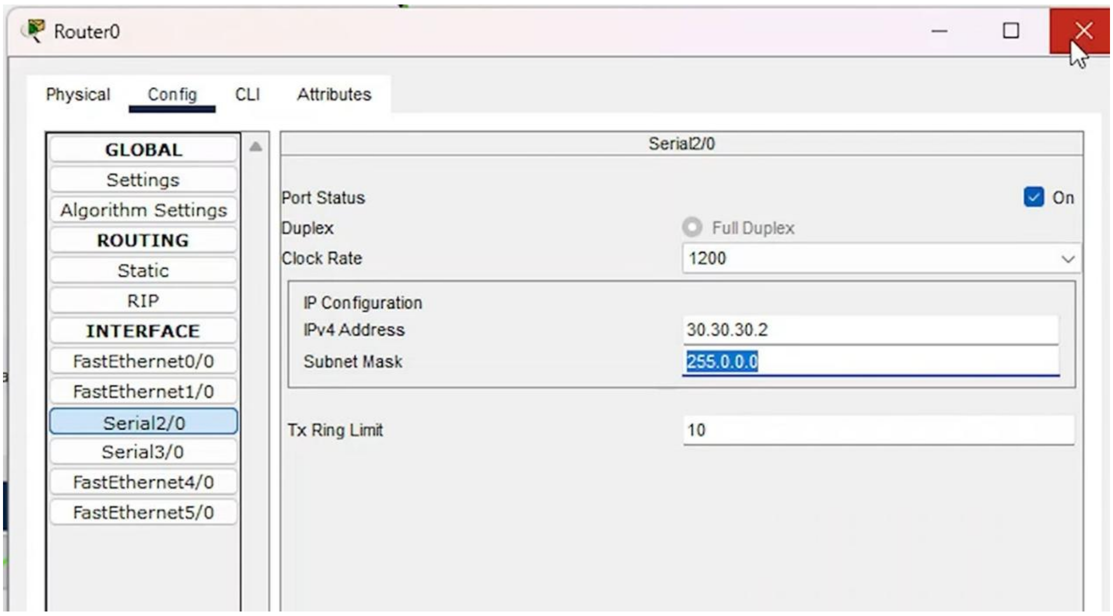


ii. Router 0 (Public Network Side):

Go to the Config tab > Interface f0/0. Turn on Port Status, assign the IPv4 address 20.20.20.0 (which is the default gateway), and set the subnet mask.



Turn on the Serial2/0 interface, and allocate IPv4 address 30.30.30.2, subnet mask 255.0.0.0.



b. Configure Static NAT and Routing on Router 1

Open the CLI interface of Router 1 and enter the following core configuration commands to establish static mappings:

Router(config-if)# **exit**

Router(config)#**ip nat inside source static 10.10.10.1 30.30.30.1**

Router(config)#**ip nat inside source static 10.10.10.2 30.30.30.1**

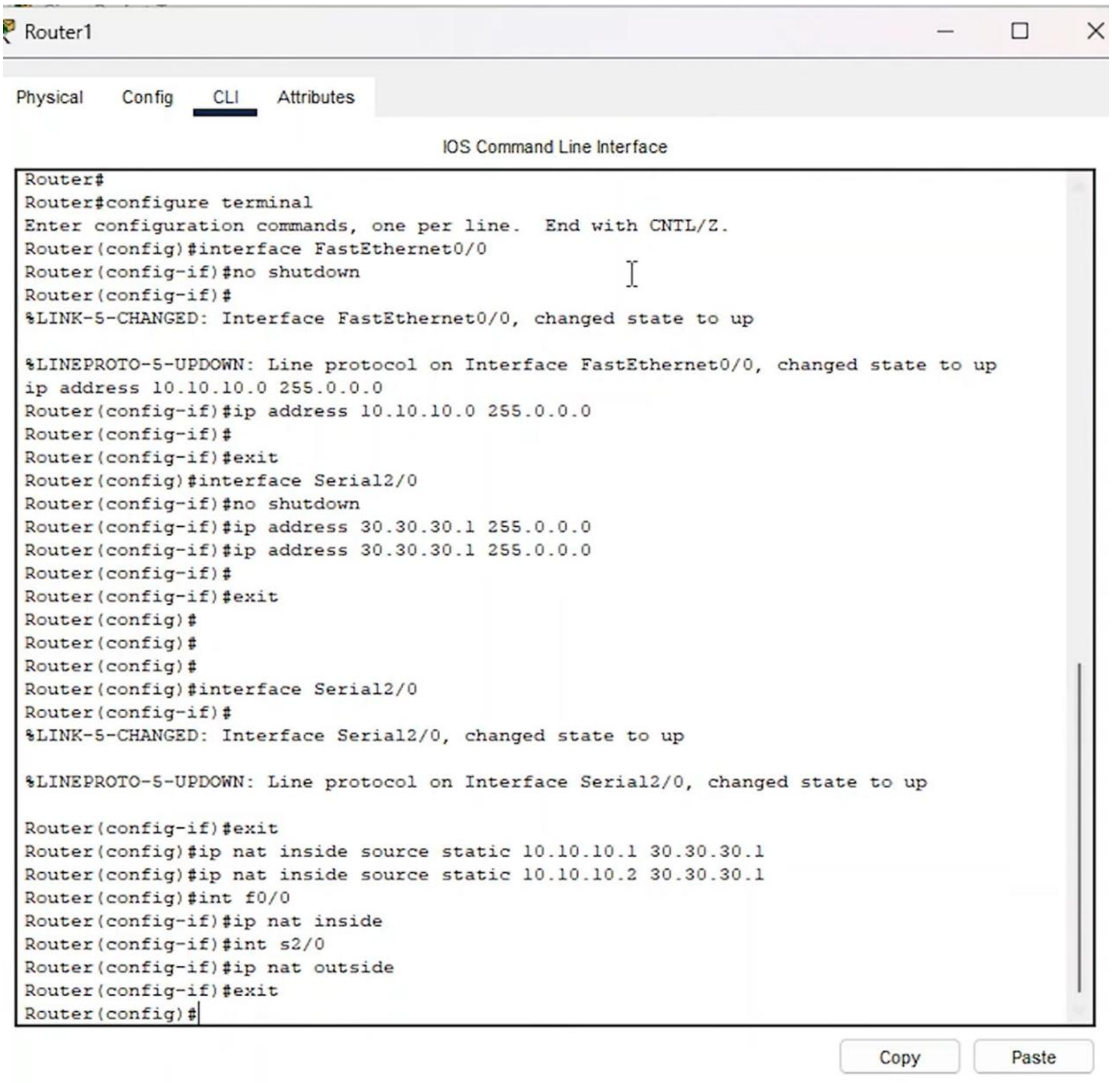
Router(config)#**int f0/0**

Router(config-if)#**ip nat inside**

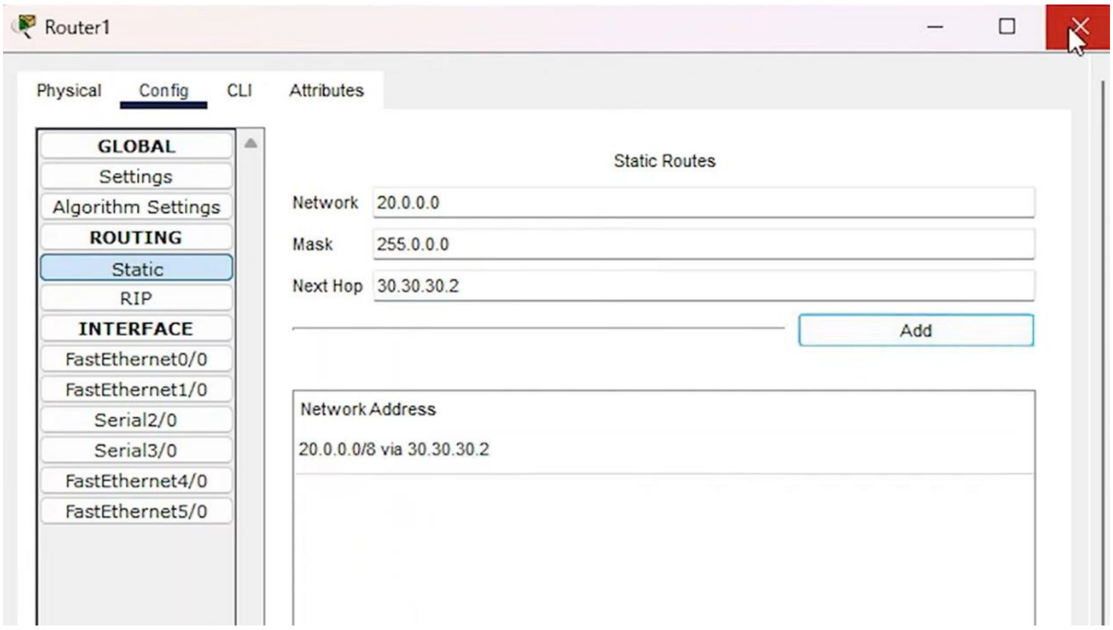
Router(config-if)#**int s2/0**

Router(config-if)#**ip nat outside**

Router(config-if)#**exit**



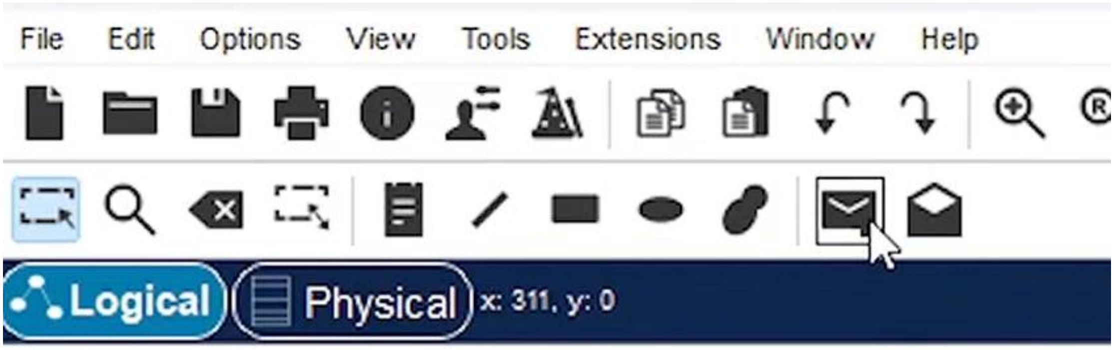
Configure Static NAT: Go to the Config tab > Static. Add the public network's IP range (20.0.0.0), subnet mask (255.0.0.0), and Next Hop address (30.30.30.2), then click Add.

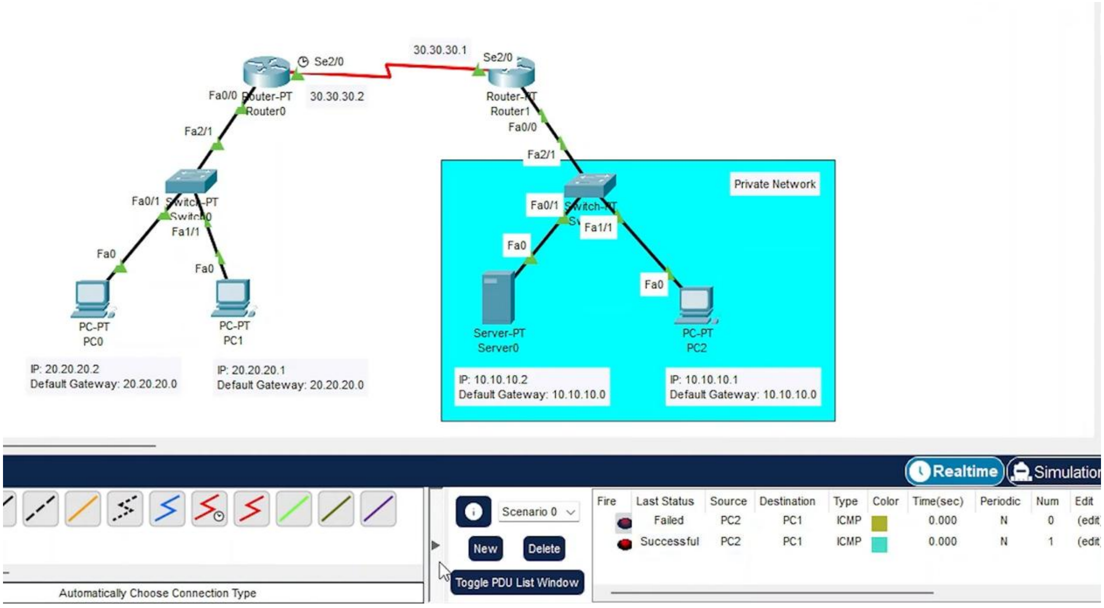


5. Experiment 3 -- Packet Testing, Ping Communication, and Web Testing

a. Connectivity Test (Ping Test)

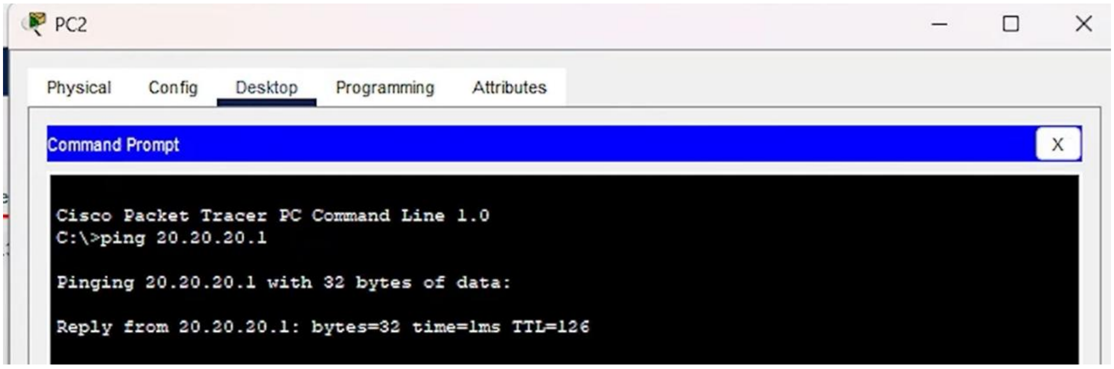
i. Select the packet icon on the toolbar, click PC2 then PC1, which means sending a packet from the private network (PC2) to the public network (PC1). The first attempt will fail, but the second attempt will succeed (as shown in the bottom right corner).



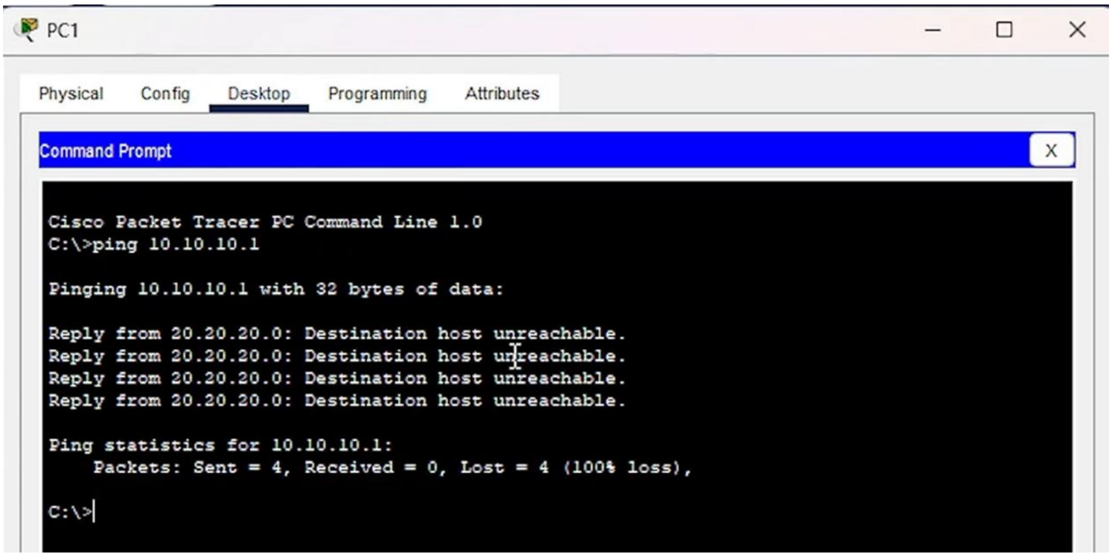


ii. Select the packet icon on the toolbar, click PC1 then PC2, sending a packet from the public network to the private network. Packets sent from the public network to the private network will always fail (because the public network cannot actively access the private network).

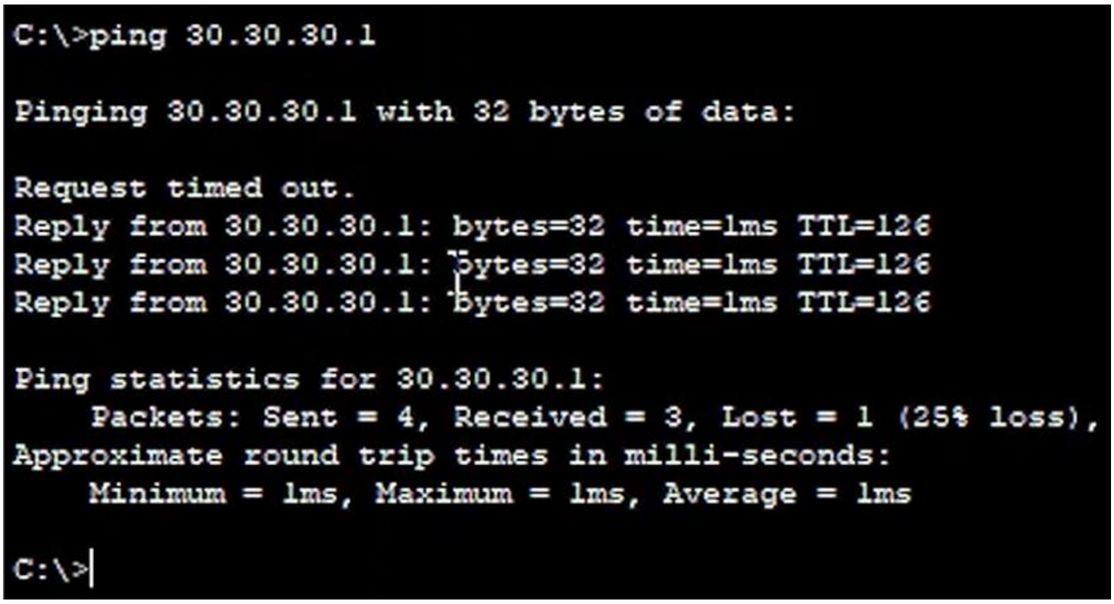
iii. Ping public network from private network: Open the Command Prompt of PC2 and execute ping 20.20.20.1. You should successfully receive a reply.



iv. Ping private network from public network: Open the Command Prompt of PC1 and execute ping 10.10.10.1. The result will show "Destination host unreachable", verifying the one-way protection of NAT.

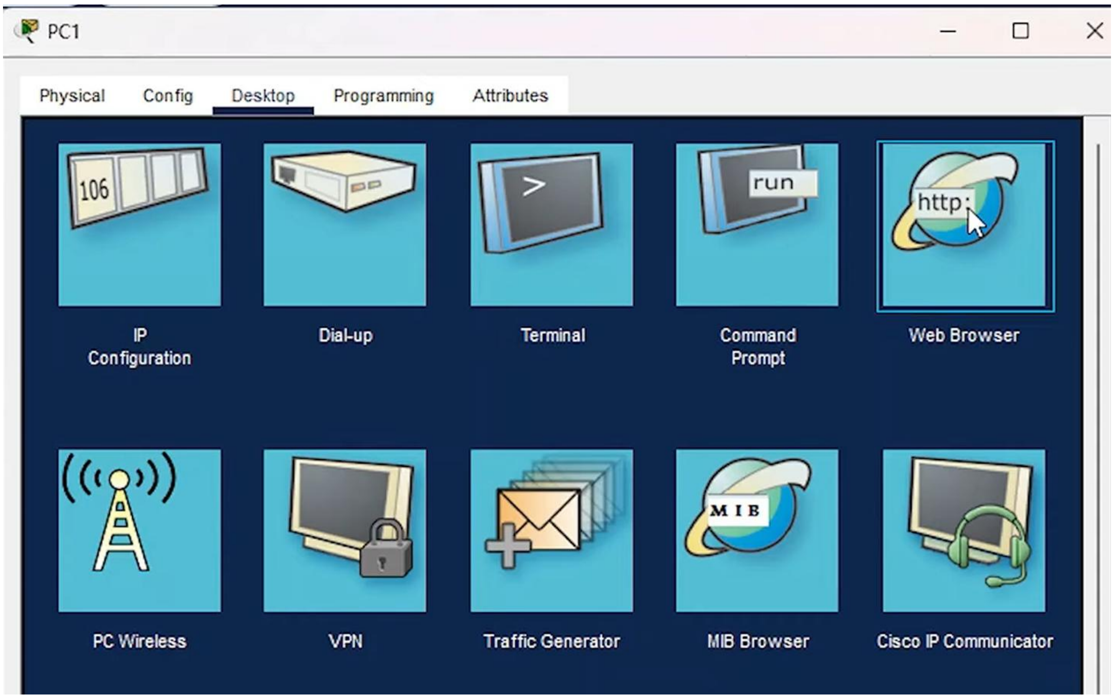


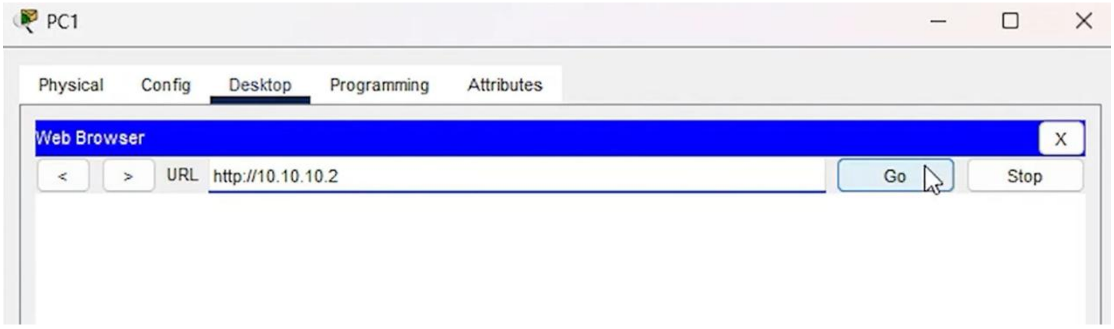
v. Ping the public mapped IP: Execute ping 30.30.30.1 from PC1, and you should successfully receive a reply.



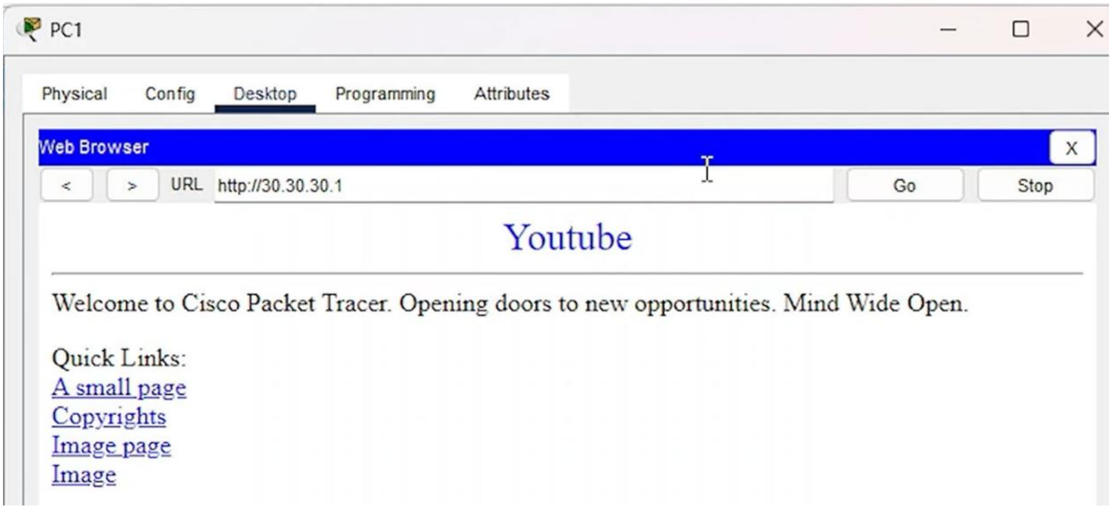
b. Web Application Layer Test (Web Test)

i. Open the Web Browser on the public network PC1. Attempt to enter the server's private IP http://10.10.10.2; the web page will be inaccessible (Request Timeout).



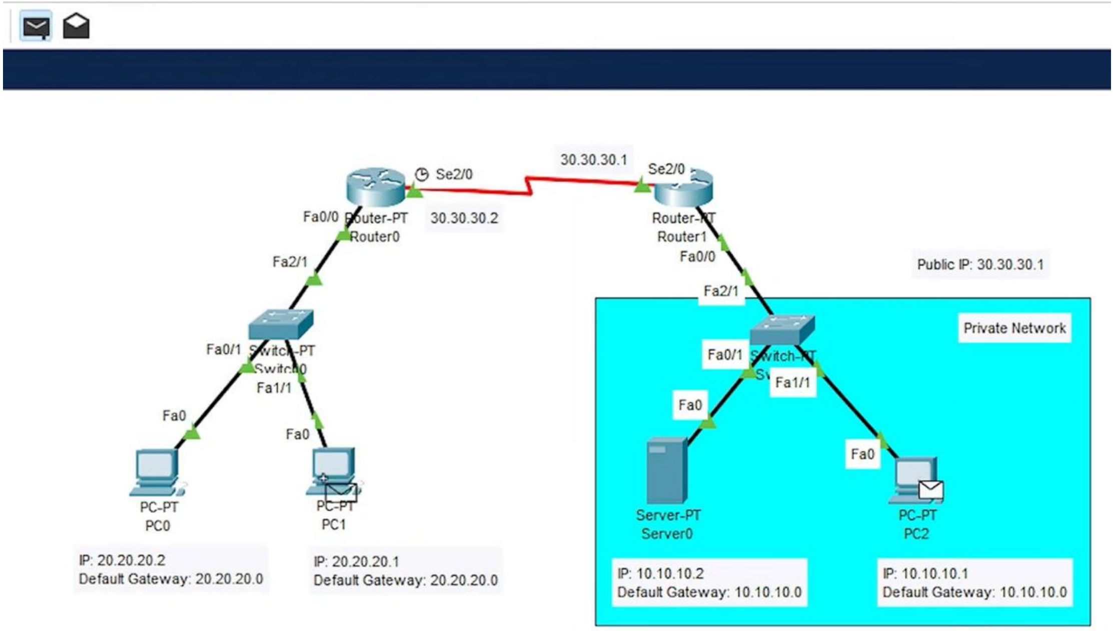


ii. Enter the mapped public IP http://30.30.30.1. The server's web page should load successfully and display the word "Youtube".

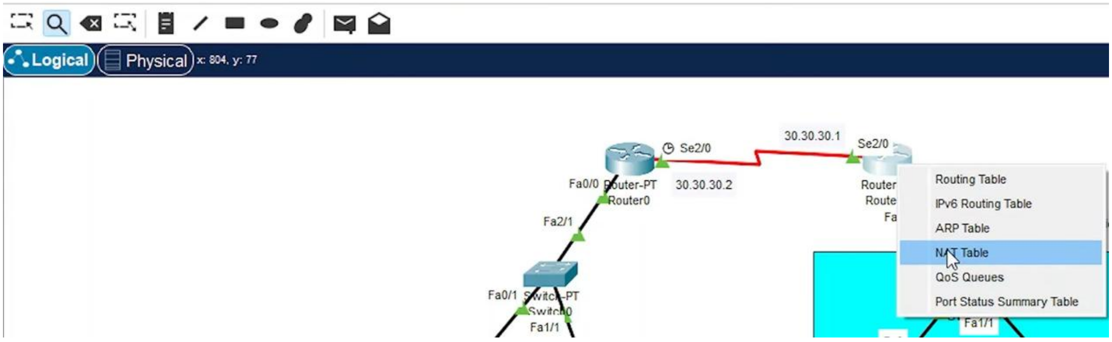


c. Check NAT Table

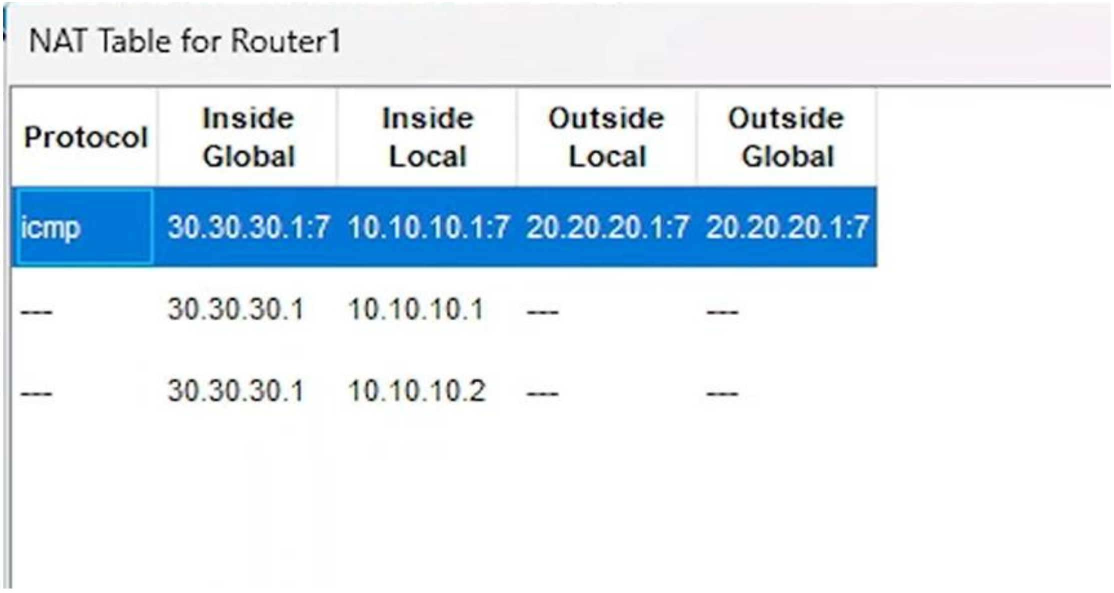
i. Close the window and send a packet from PC2 to PC1. (Refer to Step 5) .



ii. Use the Inspect tool to click on Router 1, and then view the "NAT Table".



iii. Verify the entries under "Inside Global" and "Inside Local" to confirm that the private IP addresses have been successfully mapped to the public IP address.



NAT Table for Router1

Protocol	Inside Global	Inside Local	Outside Local	Outside Global
icmp	30.30.30.1:7	10.10.10.1:7	20.20.20.1:7	20.20.20.1:7
---	30.30.30.1	10.10.10.1	---	---
---	30.30.30.1	10.10.10.2	---	---

d. Supplementary Practice & Advanced Exploration

In the basic experiment above, we verified simple static NAT. Now suppose our network scale expands: the private network side is no longer just a few computers, but includes three core schools of the university (e.g., EEE, NIE, and CCDS), and these three schools are distributed in different subnets (VLANs). However, the school still only has one external public IP address.

i. If all terminals in the three schools are required to access the internet simultaneously, is the "Static NAT" used in the experiment still applicable? If not, what technology should be introduced?

Static NAT (one-to-one mapping) is no longer applicable because it will greatly consume public IP resources. Port Address Translation (PAT), or Dynamic NAT Overload, should be introduced. PAT allows the router to allocate different Port Numbers, letting hundreds or thousands of internal private IPs share the same public IP address.

ii. If PAT is deployed on the router, what changes need to be made to the core configuration command logic?

You will no longer use ip nat inside source static to specify IPs one by one. You need to: Configure an Access Control List (ACL) to uniformly capture the private network segments of the three schools (e.g., access-list 1 permit 10.0.0.0 0.255.255.255).

Use the command with the overload keyword to dynamically overload the private network traffic bound by the ACL onto the external interface (e.g., ip nat inside source list 1 interface Serial2/0 overload).

When checking the NAT Table again, in addition to the IP address mappings, you can also clearly see the different external port numbers assigned to different terminals.